

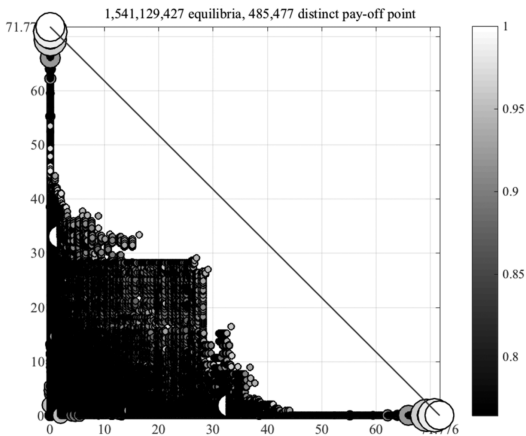
Structural Estimation of Directional Dynamic Games with Multiple Equilibria

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Multiplicity of Equilibria



Equilibrium payoff set in Bertrand pricing and investment game

1,541,129,427 MPE

Directionality $\implies$ Full solution $\implies$ Nested MLE

- ▶ Focus on the subclass of discrete stochastic games with **directional state transitions** $\rightarrow$ directional dynamic games
- ▶ **State recursion** algorithm = generalization of backward induction to solve for MPE in stages
- ▶ Multiple solutions at stage games $\rightarrow$ **RLS algorithm** to enumerate all combinations of solutions $\iff$ solve for all MPEs (**IRS, 2016**)
- ▶ Current work: develop a **nested full solution MLE estimator** $\equiv$ **NRLS estimator** based on integer programming branch-n-bound algorithm
 - ▶ Robust to **multiplicity of equilibria in the model and the data**
 - ▶ Computationally feasible
 - ▶ Computational burden decreases with sample size
- ▶ Extensive Monte Carlo study of existing estimators: 2-step, NPL, EPL, MPEC vs NRLS



Iskhakov, Rust and Schjerning (2016, ReStud)

Recursive Lexicographical Search: Finding All Markov Perfect Equilibria of Finite State Directional Dynamic Games

MLE for dynamic games with multiple equilibria

- ▶ Data from M independent markets from T periods, N players

$$\mathcal{Z} = \{a^{ijt}, x^{it}\}_{i \in \{1 \dots M\}, j \in \{1, \dots, N\}, t \in \{1, \dots, T\}}$$

- ▶ MPE is a pair of **strategy profiles** and **value functions** such that

$$V_{\theta} = \Psi^{V_{\theta}}(V_{\theta}, P_{\theta}, \theta) \quad (\text{Bellman equations})$$

$$P_{\theta} = \Psi^{P_{\theta}}(V_{\theta}, P_{\theta}, \theta) \quad (\text{mutual-best-response conditions})$$

- ▶ **Multiplicity** $\rightarrow$ set of equilibria $\mathcal{E}(\theta) = \{V_{\theta}^k, P_{\theta}^k\}_{k \in \{1, \dots, K(\theta)\}}$
- ▶ MLE estimator $\hat{\theta}^{ML}$ is given by

$$\hat{\theta}^{ML} = \arg \max_{\theta} \left[\max_{k \in \{1, \dots, K(\theta)\}} \sum_{t=1}^T \sum_{i=1}^M \sum_{j=1}^N \log P_j^k(a^{ijt} | x^{it}; \theta) \right]$$

- ▶ One equilibrium in the data, relaxed later using known groups of markets
- ▶ Inner loop requires full solution, impossible?

MLE by constrained optimization (MPEC)

- ▶ Idea: use discretized values of P and V as *variables*
- ▶ **Augmented log-likelihood** function

$$\mathcal{L}(Z, P, \theta) = \sum_{t=1}^T \sum_{i=1}^M \sum_{j=1}^N \log P_j(a^{ijt} | x^{it}; \theta)$$

- ▶ The constrained optimization formulation of the ML estimation problem is

$$\max_{\theta, P, V} \mathcal{L}(Z, P, \theta) \quad \text{subject to} \quad \begin{cases} V = \Psi^V(V, P, \theta) \\ P = \Psi^P(V, P, \theta) \end{cases}$$

- ▶ May work with multiple equilibria with smart optimization algorithms
- ▶ **Much bigger computational problem**
- ▶ Same MLE only at the global constrained solution



Su (2013); Egesdal, Lai and Su (2015)

Other existing estimation methods

▶ Two step (CCP) estimators

- ▶ Fast, do not impose equilibrium constraints, finite sample bias

1. Estimate CCP $\rightarrow \hat{P}$
2. Method of moments • Minimal distance • Pseudo likelihood



Hotz, Miller (1993); Altug, Miller (1998); Pakes, Ostrovsky, and Berry (2007); Pesendorfer, Schmidt-Dengler (2008)

▶ Nested pseudo-likelihood (NPL)

- ▶ Recursive two step pseudo-likelihood
- ▶ Bridges the gap between efficiency and tractability
- ▶ Unstable under multiplicity



Aguirregabiria, Mira (2007); Aguirregabiria, Marcoux (2021)

▶ Efficient pseudo-likelihood (EPL)

- ▶ Incorporates Newton step in the NPL operator
- ▶ More robust to the stability and multiplicity of equilibria



Dearing, Blevins (2024)

Dynamic Bertrand price competition

Directional stochastic dynamic game

- ▶ Two Bertrand competitors, $n = 2$, no entry or exit
- ▶ Discrete time, infinite horizon ($t = 1, 2, \dots, \infty$)
- ▶ Firms maximize expected discounted profits
- ▶ Each firm has two choices in each period:
 1. Price for the product — simultaneous
 2. Whether or not to buy the state of the art technology
 - ▶ Simultaneous moves
 - ▶ Alternating moves

Static Bertrand price competition in each period

- ▶ Continuum of consumers make static purchase decision
- ▶ No switching costs: buy from the lower price supplier
- ▶ Per period profits (c_i is the marginal cost)

$$r_i(c_1, c_2) = \begin{cases} 0 & \text{if } c_i \geq c_j \\ c_j - c_i & \text{if } c_i < c_j \end{cases}$$

Cost-reducing investments

State-of-the-art production cost c process

- ▶ Initial value c_0 , lowest value 0: $0 \leq c \leq c_0$
- ▶ Discretized with n points
- ▶ Follows exogenous Markov process and only improves
- ▶ Markov transition probability $\pi(c_{t+1}|c_t)$
 $\pi(c_{t+1}|c_t) = 0$ if $c_{t+1} > c_t$

State space of the problem

- ▶ State of the game: cost structure (c_1, c_2, c)
- ▶ State space is $S = (c_1, c_2, c) \subset R^3$: $c_1 \geq c$, $c_2 \geq c$
- ▶ Actions are observable
- ▶ Private information EV(1) i.i.d. shocks $\eta\epsilon_{i,I}$ and $\eta\epsilon_{i,N}$

Bellman equations, firm 1, simultaneous moves

$$V_1(c_1, c_2, c, \epsilon_1) = \max [v_1(I, c_1, c_2, c) + \eta\epsilon_1(I), v_1(N, c_1, c_2, c) + \eta\epsilon_1(N)]$$

$$\begin{aligned}v_1(N, c_1, c_2, c) &= r_1(c_1, c_2) + \beta EV_1(c_1, c_2, c, N) \\v_1(I, c_1, c_2, c) &= r_1(c_1, c_2) - K(c) + \beta EV_1(c_1, c_2, c, I)\end{aligned}$$

With extreme value shocks, the investment probability (CCP) is

$$P_1(I|c_1, c_2, c) = \frac{\exp\{v_1(I, c_1, c_2, c)/\eta\}}{\exp\{v_1(I, c_1, c_2, c)/\eta\} + \exp\{v_1(N, c_1, c_2, c)/\eta\}}$$

- ▶ There is a separate Bellman equation for player 2, with “outputs” V_2 and P_2 , where $P_2(I|c_1, c_2, c)$ is firm 2’s probability of investing in state (c_1, c_2, c) .

Bellman equations, firm 1, simultaneous moves

The expected values are given by

$$EV_1(c_1, c_2, c, N) = \int_0^c \left[P_2(I|c_1, c_2, c) H_1(c_1, c, c') + [1 - P_2(I|c_1, c_2, c)] H_1(c_1, c_2, c') \right] \pi(dc'|c)$$

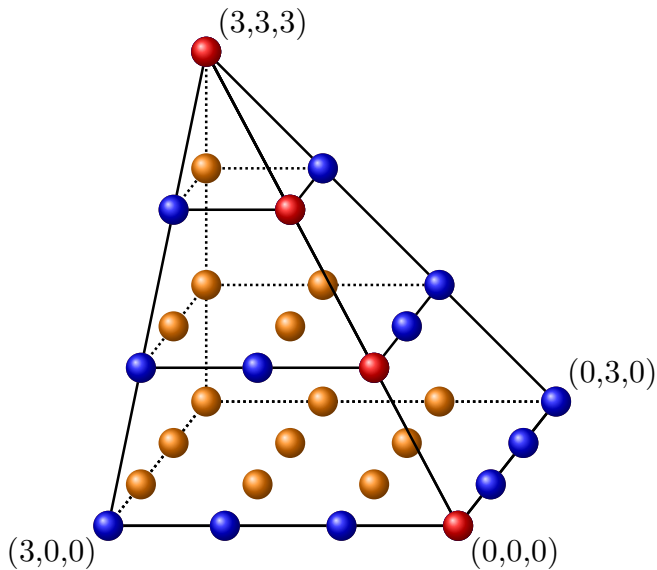
$$EV_1(c_1, c_2, c, I) = \int_0^c \left[P_2(I|c_1, c_2, c) H_1(c, c, c') + [1 - P_2(I|c_1, c_2, c)] H_1(c, c_2, c') \right] \pi(dc'|c)$$

$$H_1(c_1, c_2, c) = \eta \log \left[\exp(v_1^N(c_1, c_2, c)/\eta) + \exp(v_1^I(c_1, c_2, c)/\eta) \right].$$

is the “smoothed max” or logsum function

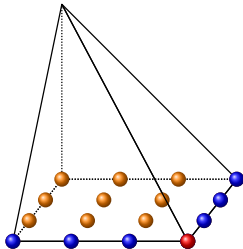
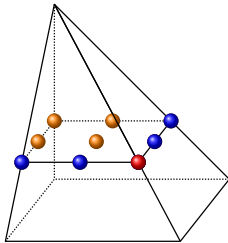
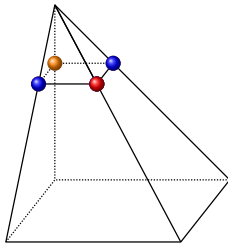
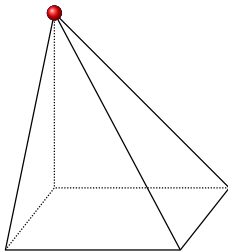
Discretized state space = a "quarter pyramid"

$$S = \{(c_1, c_2, c) | c_1 \geq c, c_2 \geq c, c \in [0, 3]\}, n = 4$$



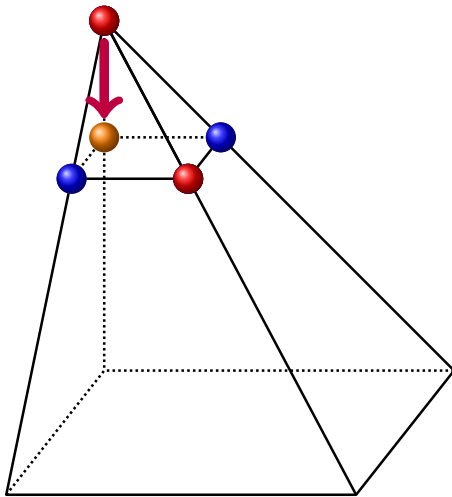
Transitions due to technological progress

As c decreases, the game falls through the layers of the pyramid



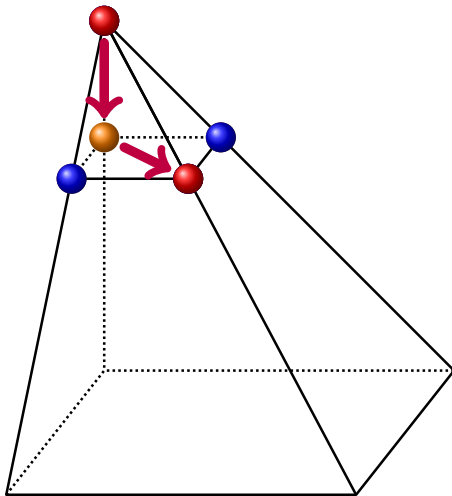
Game dynamics: example

The game starts at the apex, as some point technology improves



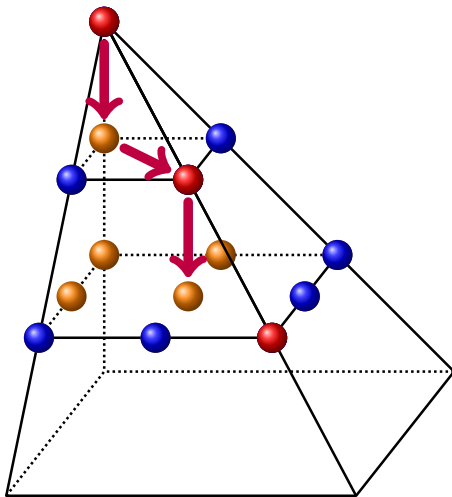
Game dynamics: example

Both firms buy new technology $c = 2 \rightsquigarrow (c_1, c_2, c) = (2, 2, 2)$



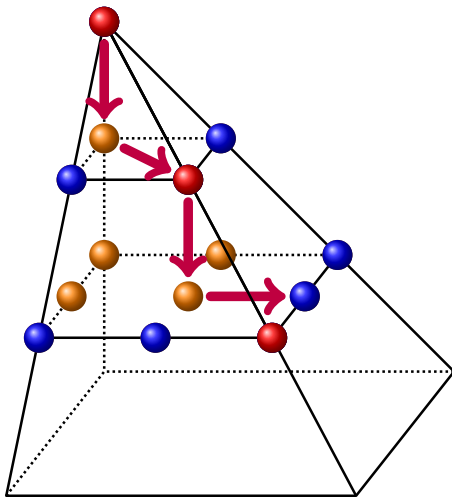
Game dynamics: example

State-of-the-art technology becomes $c = 1 \rightsquigarrow (c_1, c_2, c) = (2, 2, 1)$



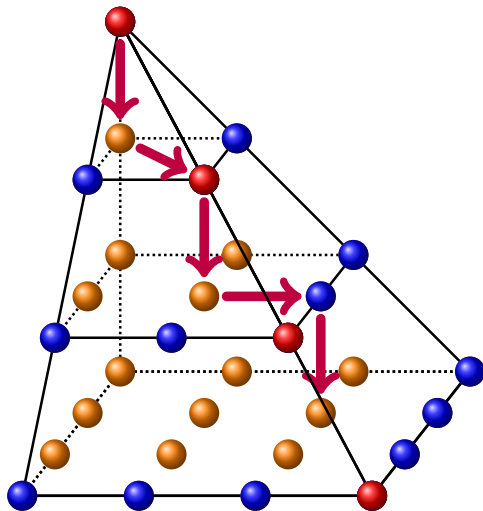
Game dynamics: example

Firm 1 invests and becomes cost leader $\rightsquigarrow (c_1, c_2, c) = (1, 2, 1)$

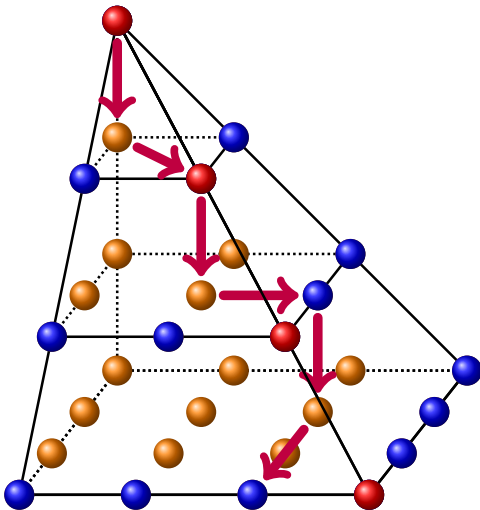


Game dynamics: example

State-of-the-art technology becomes $c = 0 \rightsquigarrow (c_1, c_2, c) = (1, 2, 0)$

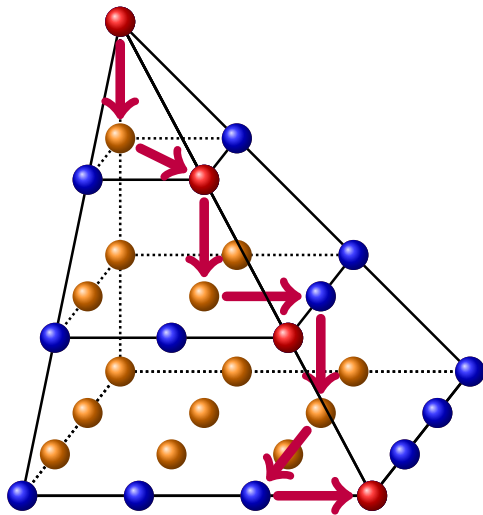


Firm 2 leapfrogs firm 1 to become new cost leader $\rightsquigarrow (c_1, c_2, c) = (1, 0, 0)$



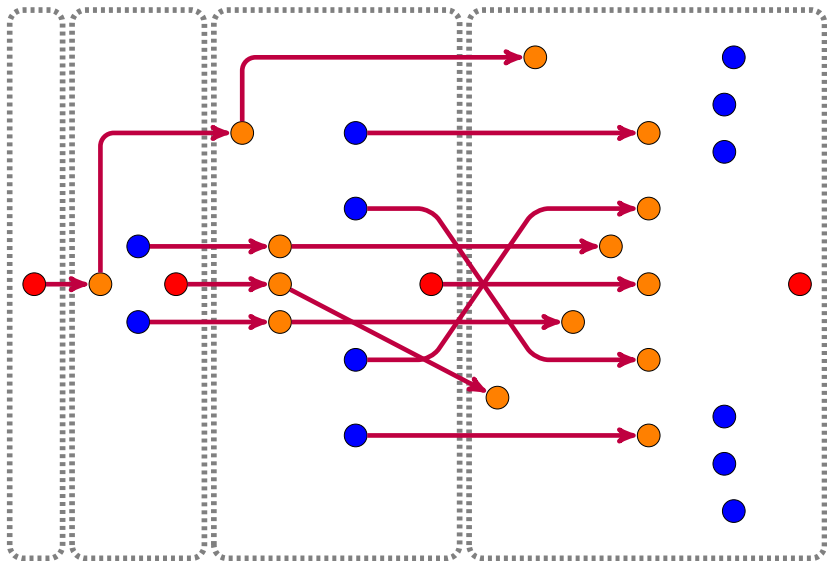
Game dynamics: example

A particular sequence of investment decisions along technological progress pass



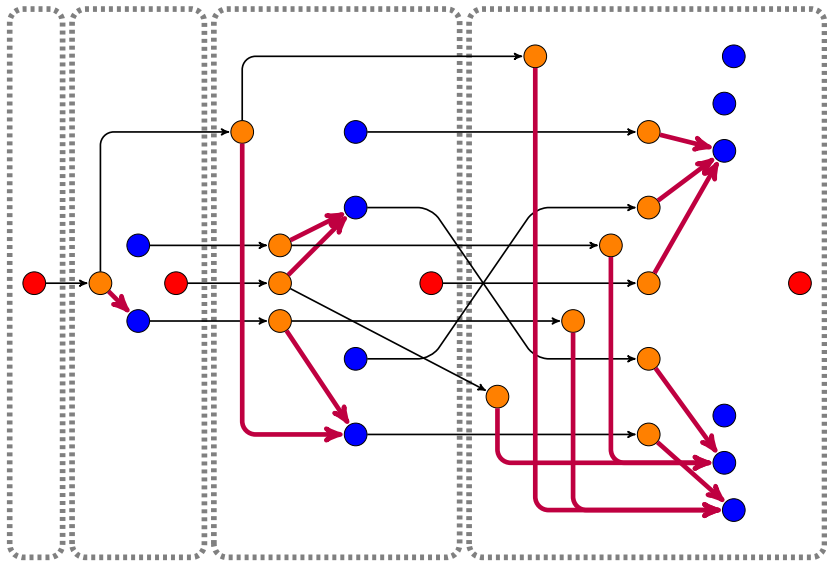
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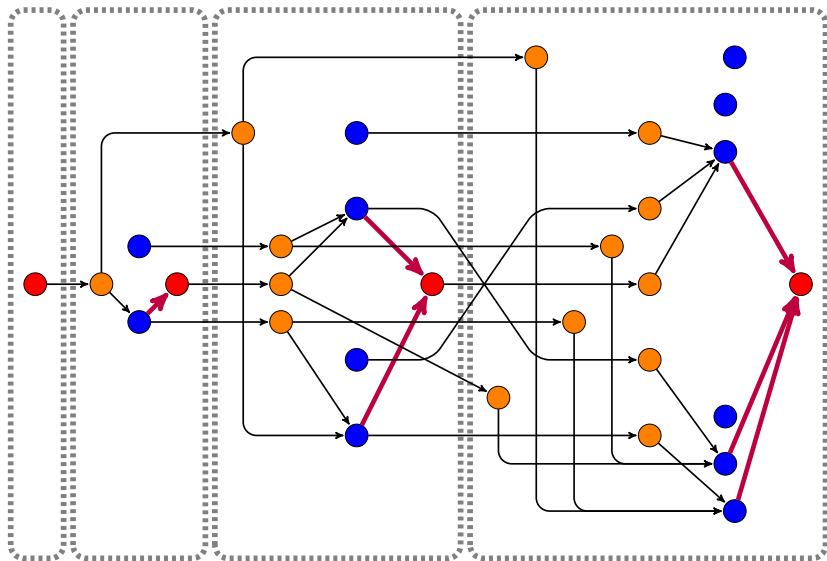
Strategy-specific partial order on S

Strategy σ_1 of firm 1: invest at all interior points



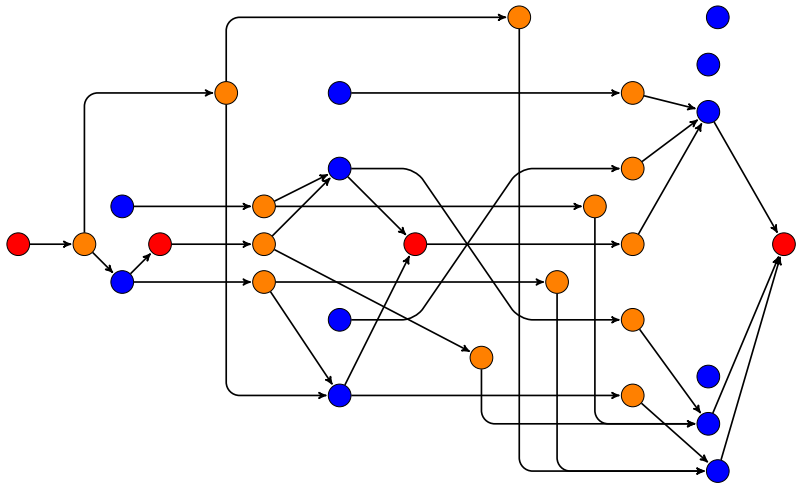
Strategy-specific partial order on S

Strategy σ_2 of firm 2: invest at all edge points



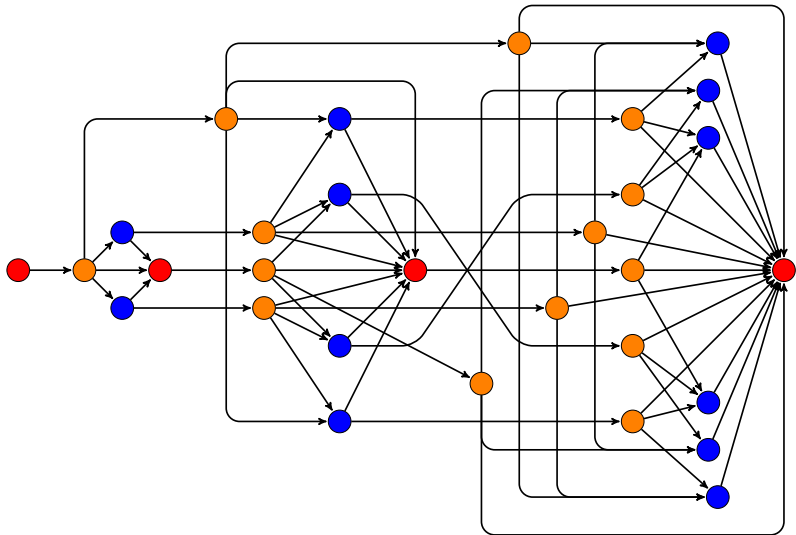
Strategy-specific partial order on S

Strategy $\sigma = (\sigma_1, \sigma_2)$ of both firms



Strategy independent partial order on S

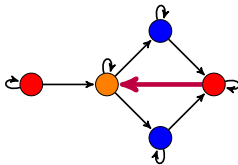
Coarsest common refinement of partial orders induced by all strategies



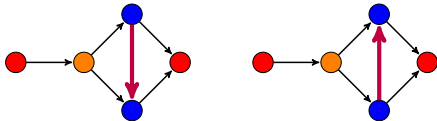
Definition of the Dynamic Directional Games

Finite state Markovian stochastic game is a DDG if it holds:

1. Every feasible Markovian strategy σ satisfies the no loop condition.



2. Every pair of feasible Markovian strategies σ and σ' induce consistent partial orders on the state space.



Iskhakov, Rust and Schjerring (2016)

Directionality can be checked from the primitives

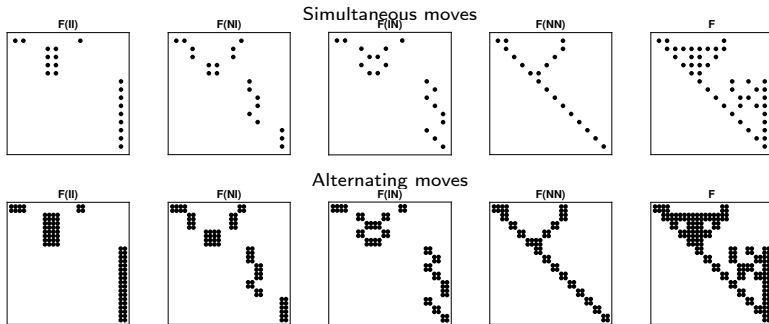
Partition $X = D \times X'$ into substages $B_d = \{d\} \times X'$. For each action profile a , let $F(a)$ be the conditional transition matrix.

The game is directional iff the **same ordering** makes every $F(a)$ upper block-triangular:

$$F(a) = \begin{pmatrix} F_{11}(a) & F_{12}(a) & \cdots & F_{1D}(a) \\ 0 & F_{22}(a) & \cdots & F_{2D}(a) \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & F_{DD}(a) \end{pmatrix} \quad \forall a.$$

- ▶ No feasible one-step transition moves to an earlier substage.
- ▶ Equivalently, the graph of all feasible transitions is acyclic.
- ▶ A **topological sort** gives the order used by state recursion.

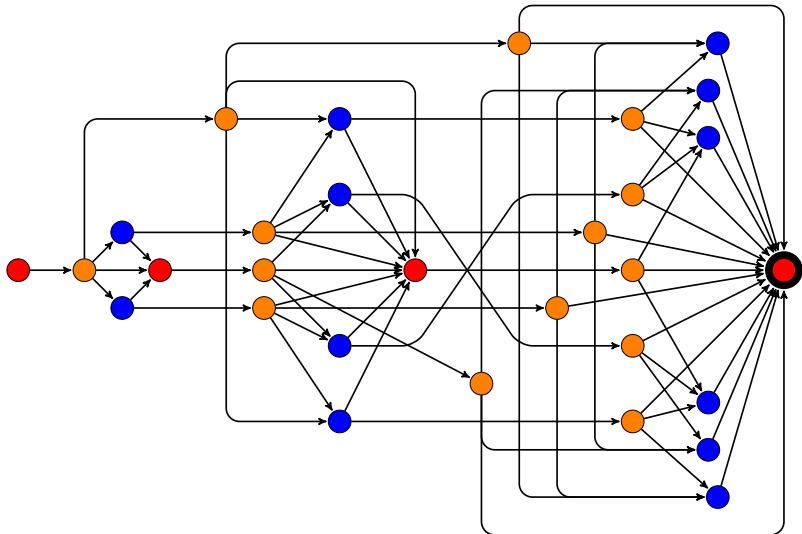
Leapfrogging transitions share a common ordering



- ▶ All conditional $F(a)$, and their union F , have common upper block-triangular support.
- ▶ With alternating moves, the non-directional mover m gives 2×2 blocks.

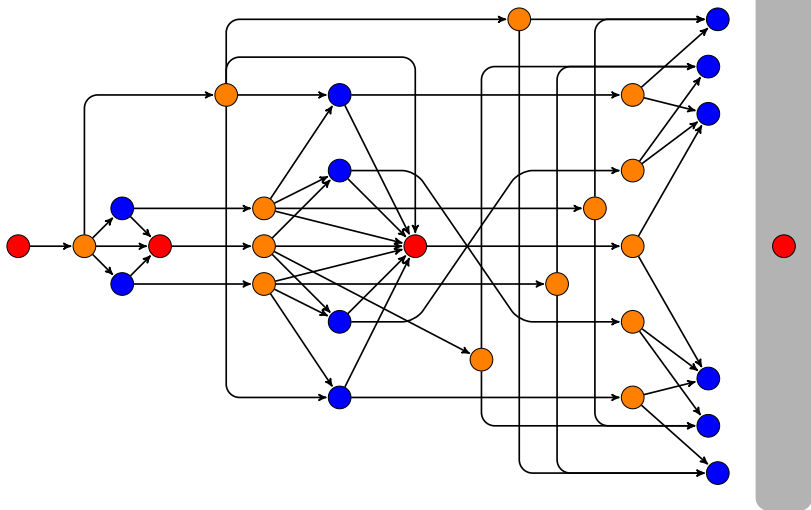
DAG recursion to partition S into stages

Identify terminal states



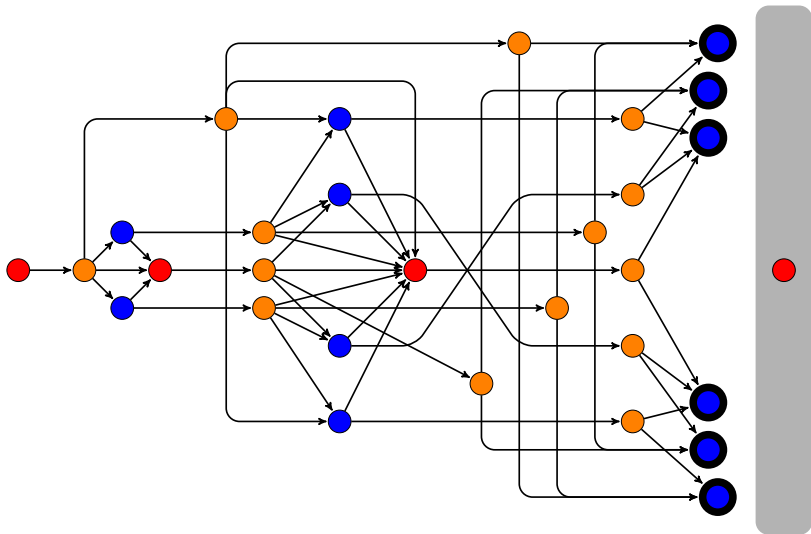
DAG recursion to partition S into stages

Remove terminal states



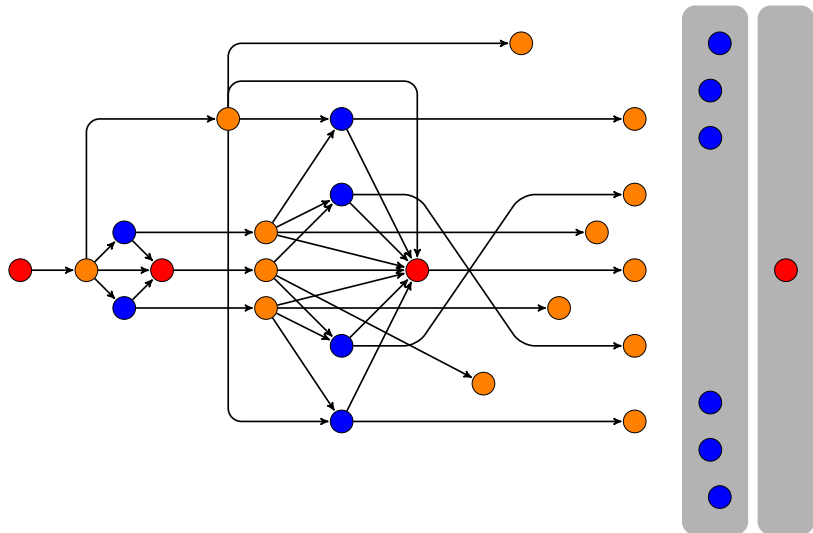
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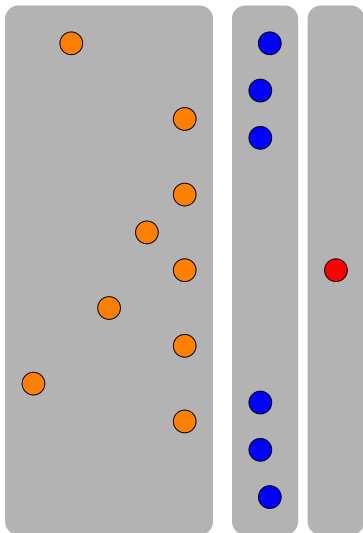
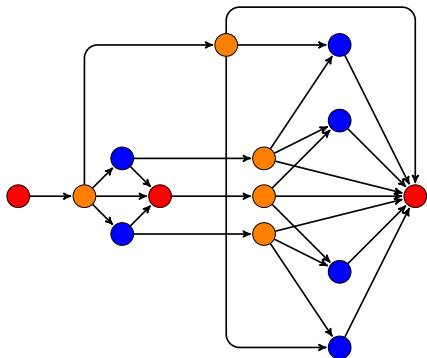
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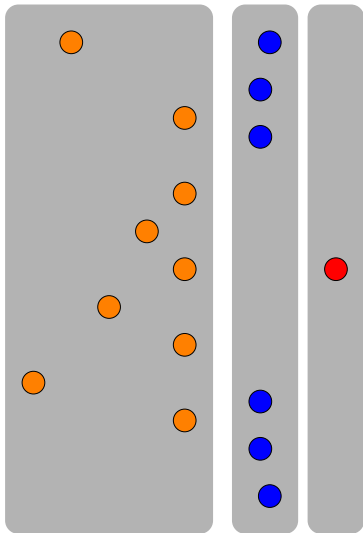
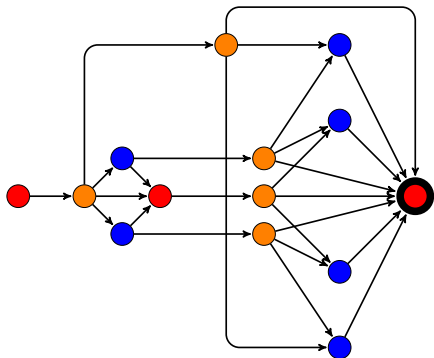
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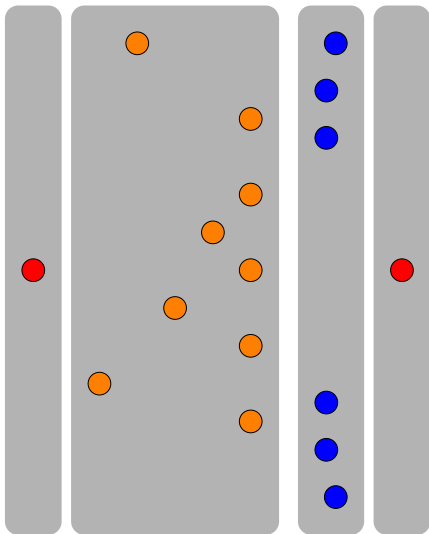
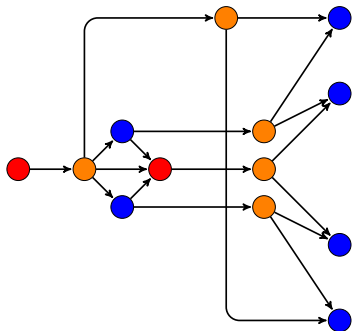
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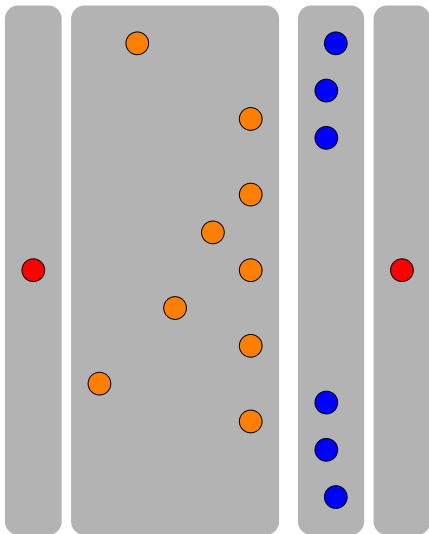
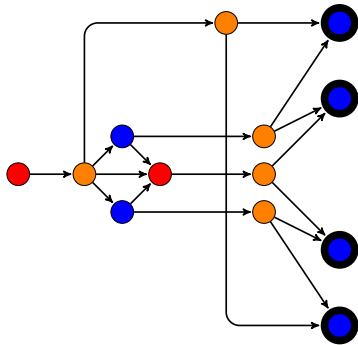
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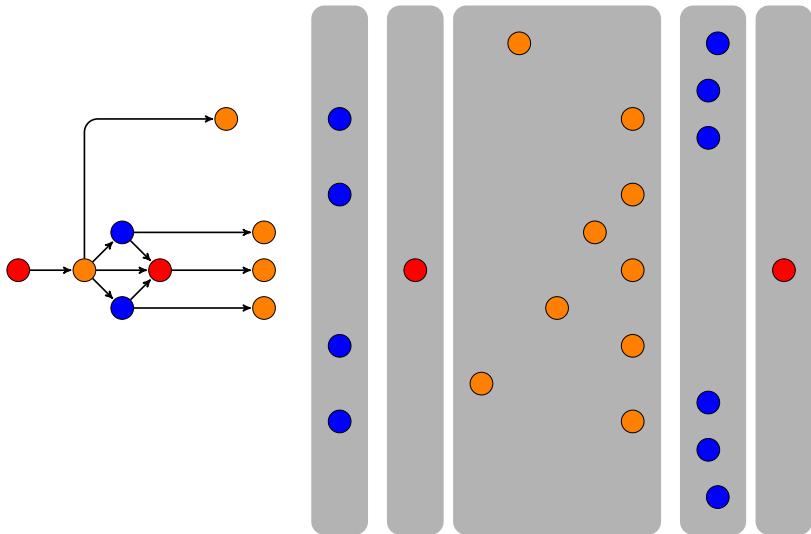
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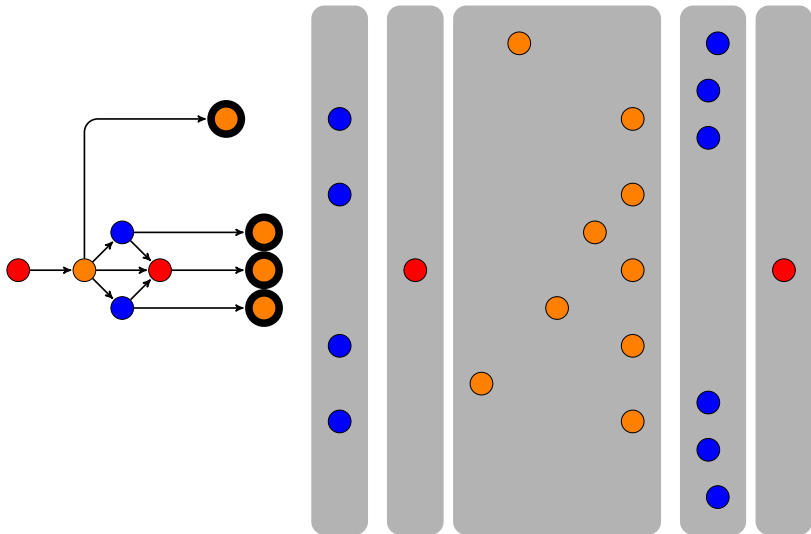
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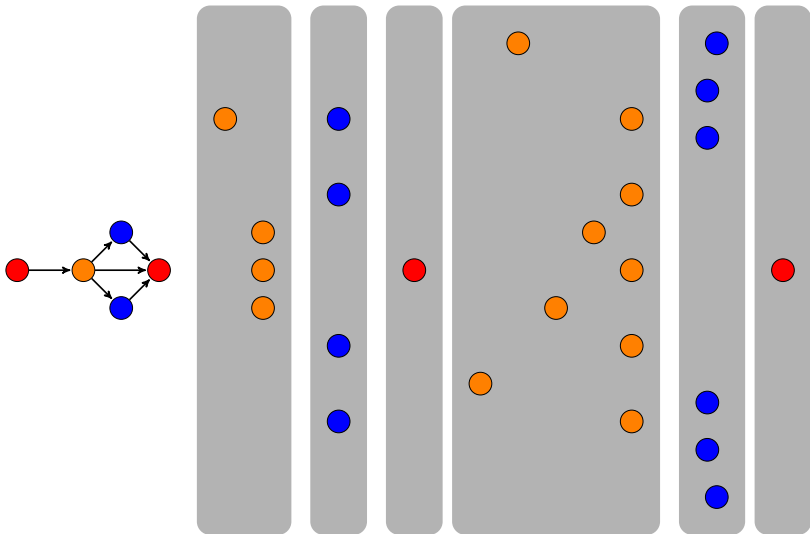
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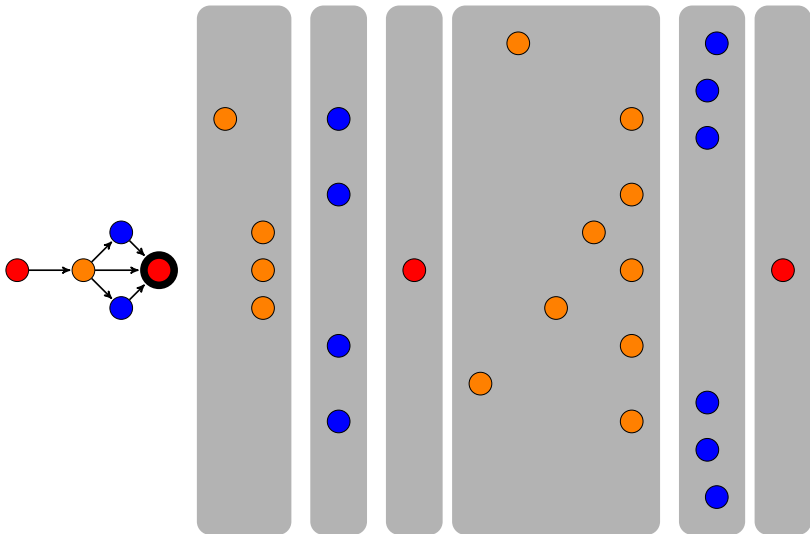
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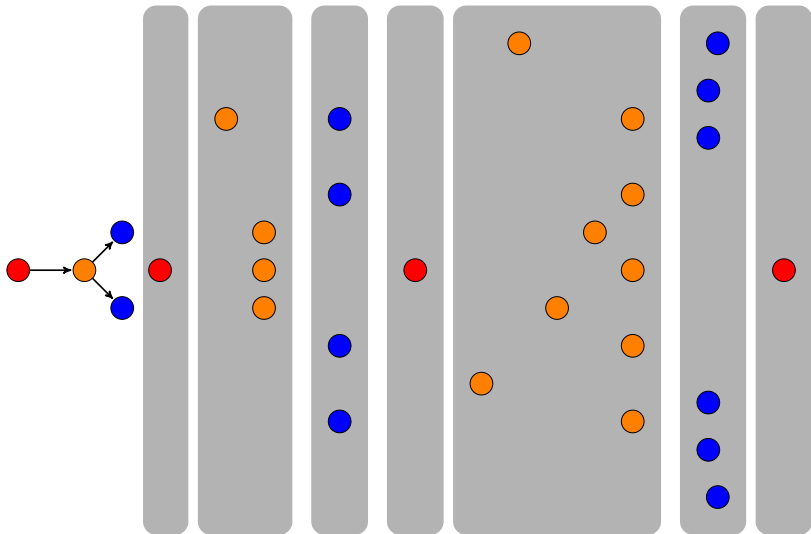
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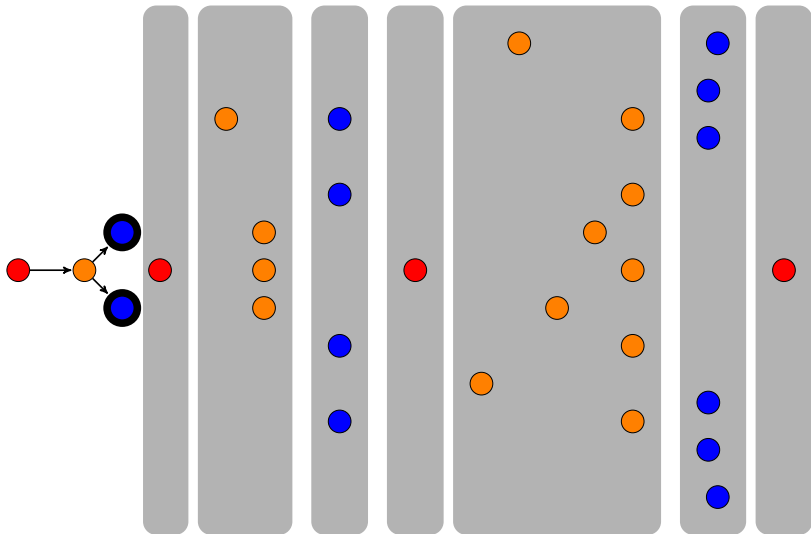
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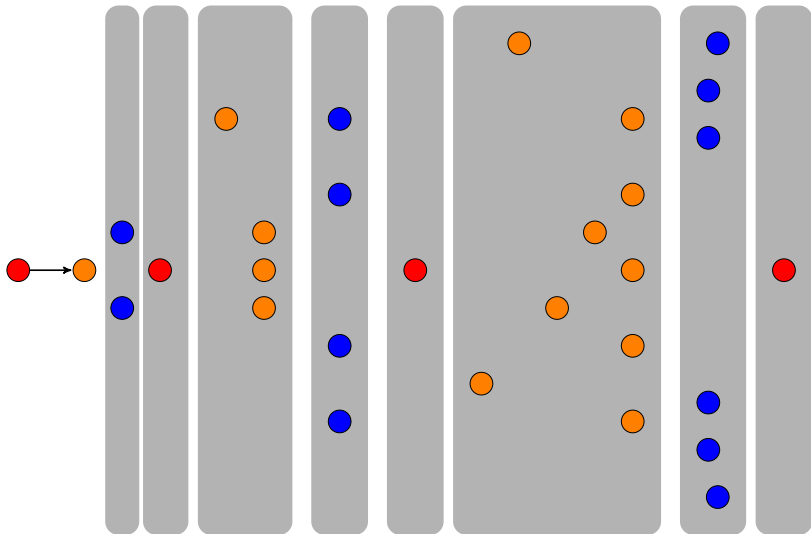
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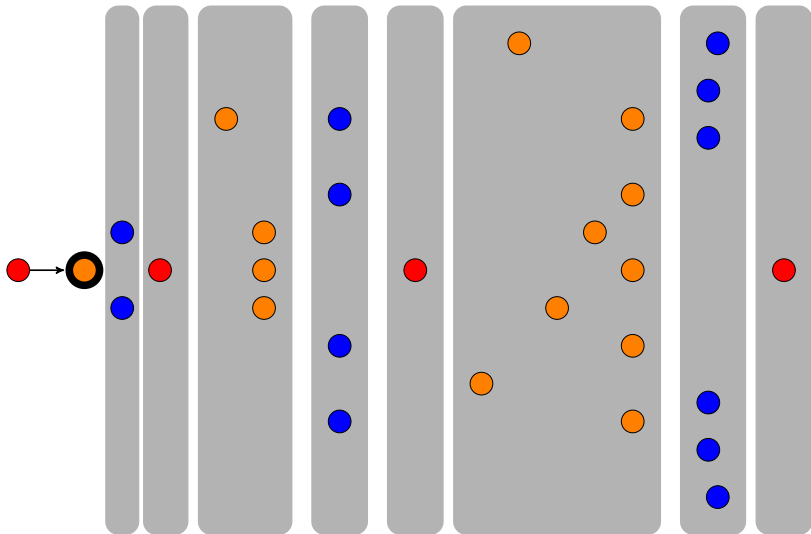
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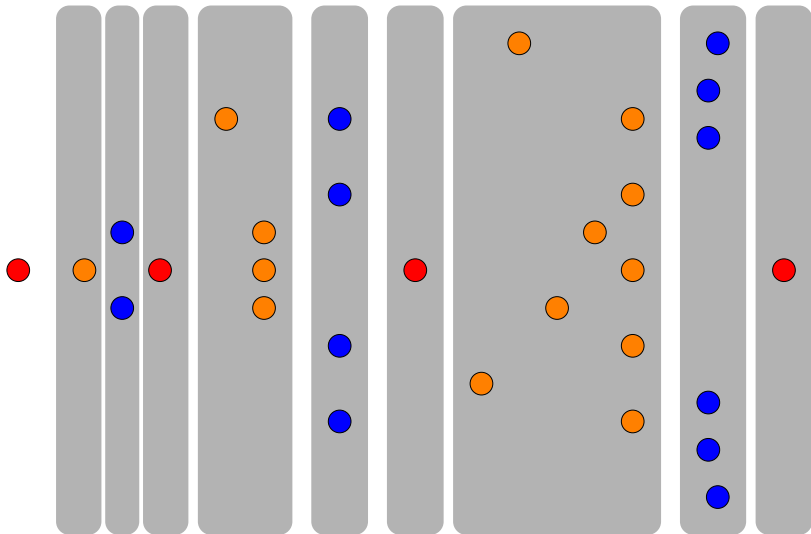
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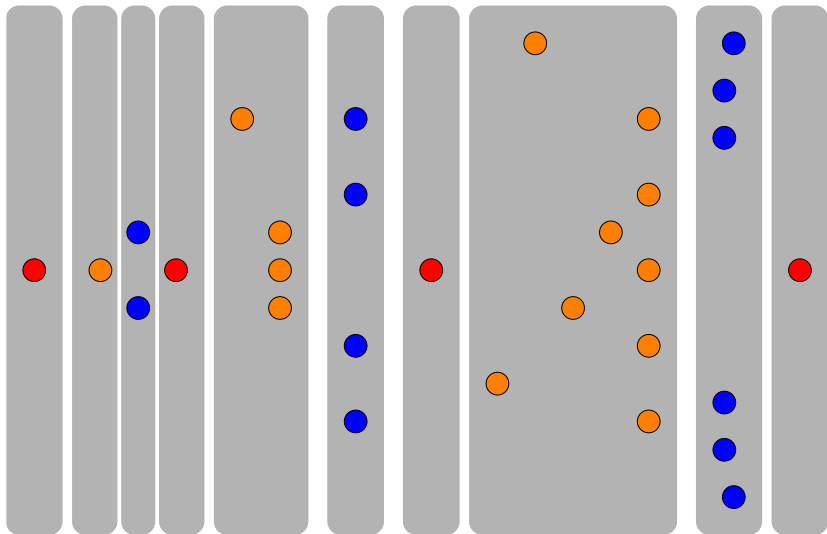
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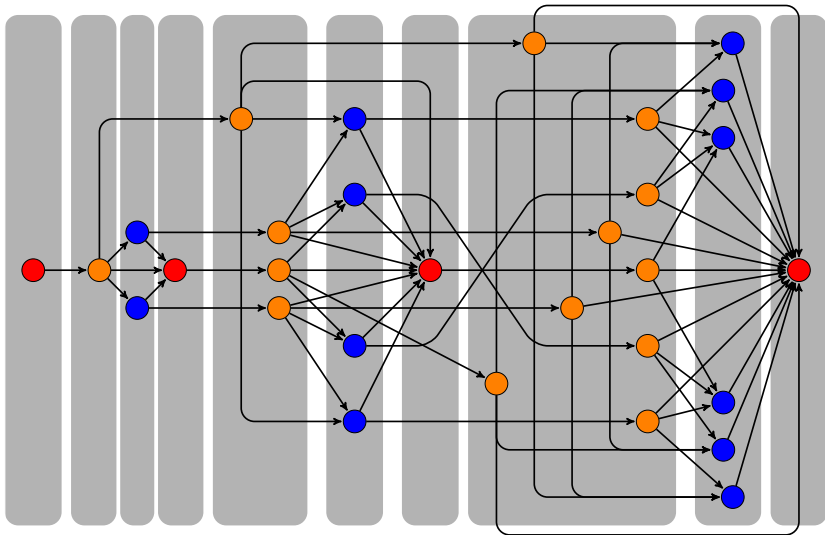
DAG recursion to partition S into stages

The stages of the game are totally ordered



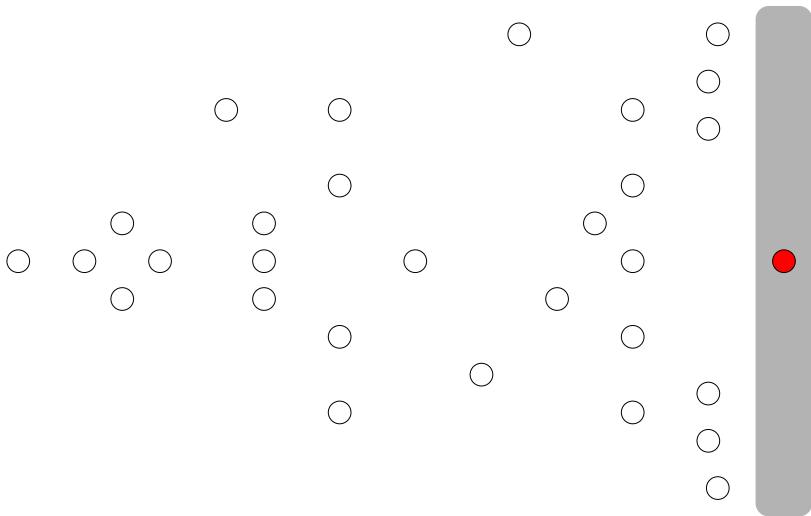
Total order on the set of stages

After running a topological sort algorithm on the DAG



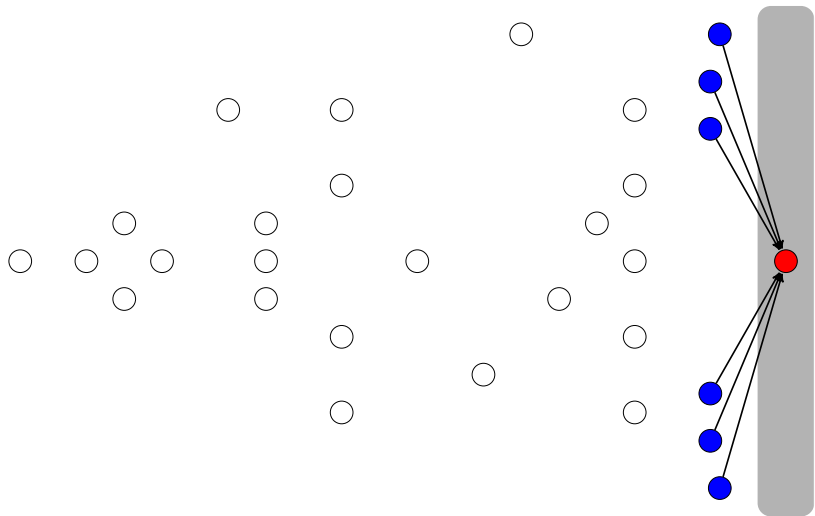
State recursion algorithm

Backward induction on stages of DDG



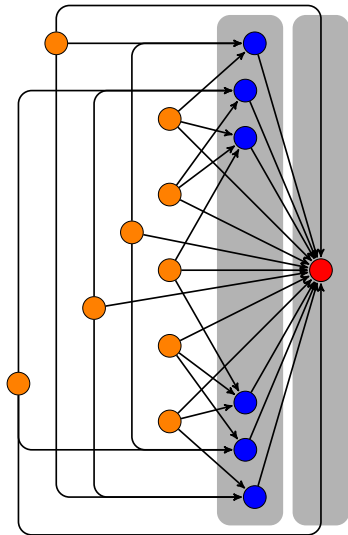
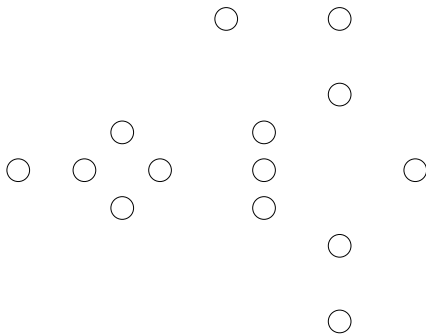
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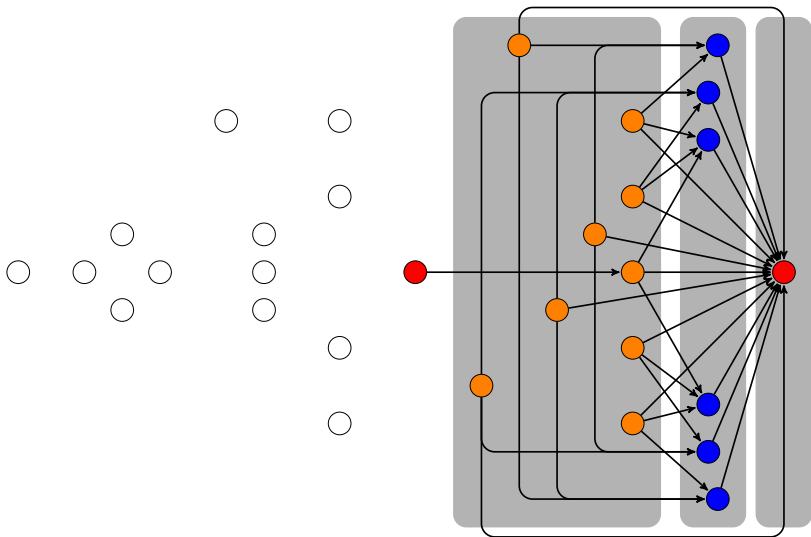
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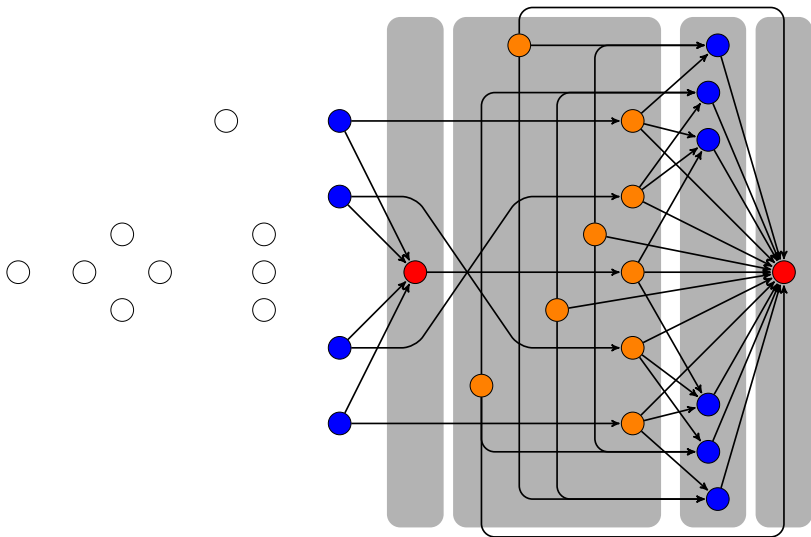
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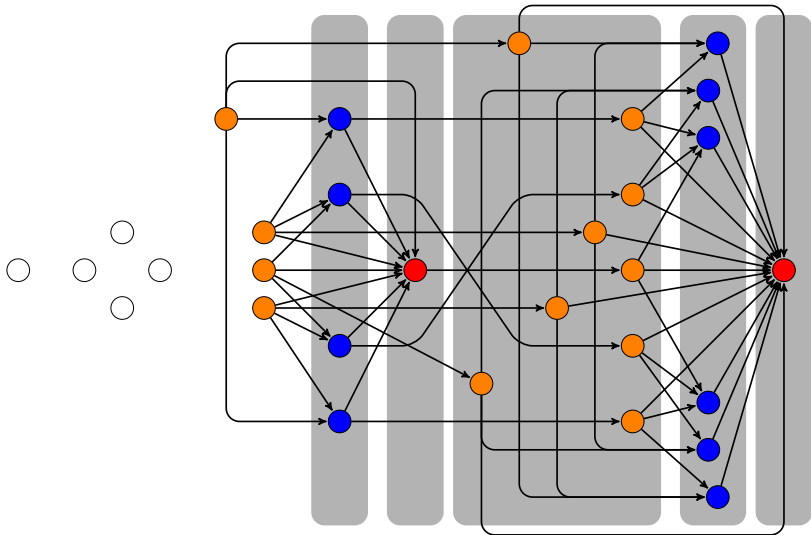
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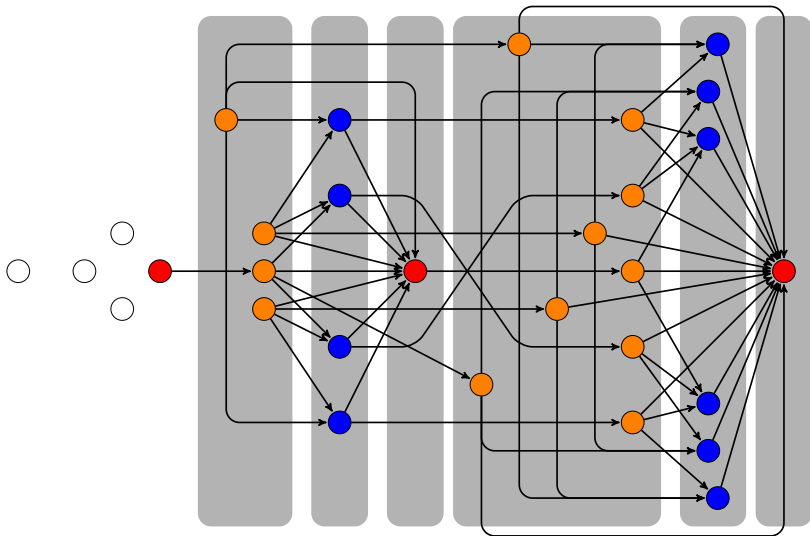
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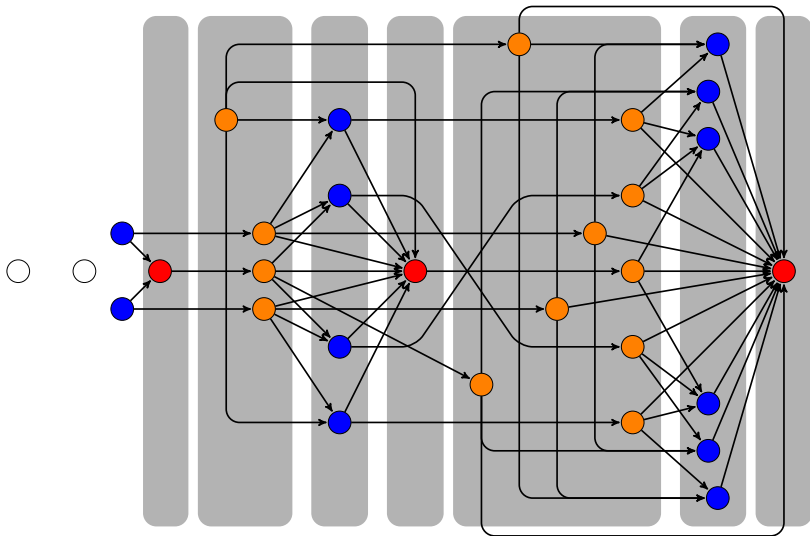
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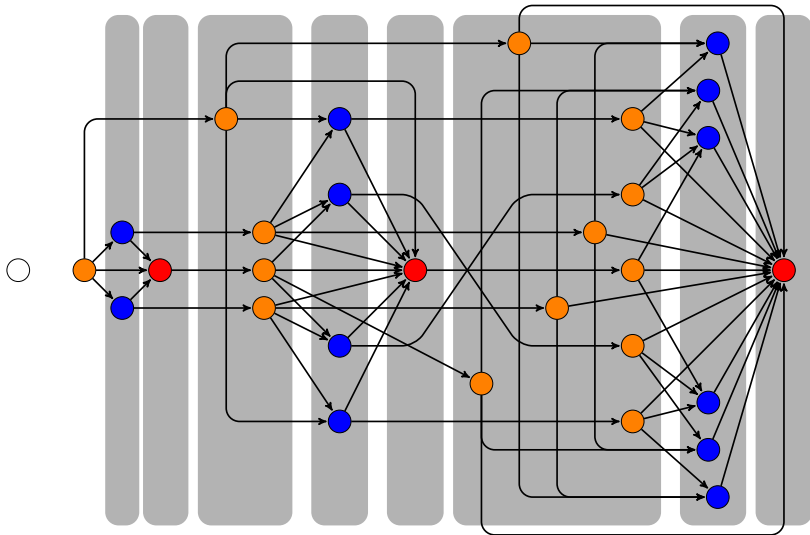
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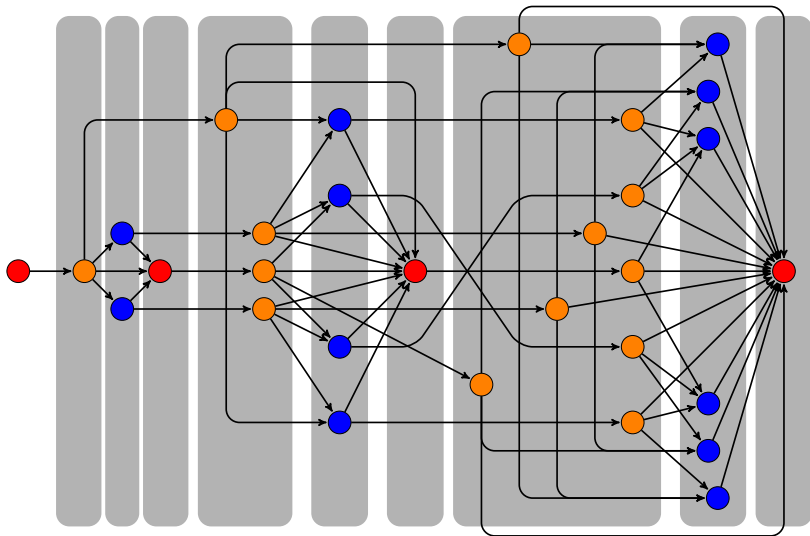
State recursion algorithm

Backward induction on stages of DDG



State recursion algorithm

Backward induction on stages of DDG

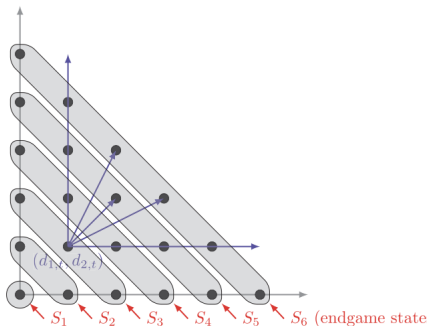
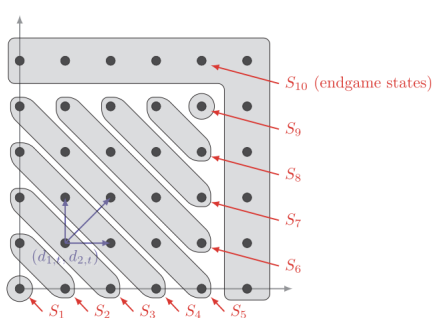


State recursion is backward induction on the state space

- ▶ Backward induction: time; state recursion: **ordered stages**.
- ▶ Transitions may stay or skip ahead, but never move backward.
- ▶ Downstream solutions fix current continuation values.
- ▶ Conditional on the selected stage equilibria.
- ▶ One selection sequence = one MPE; multiplicity motivates RLS.

Examples of Directional Dynamic Games

Many games have state dynamic evolutions described by a DAGs



Judd, Schmedders, Yeltekin (2012), *IER*

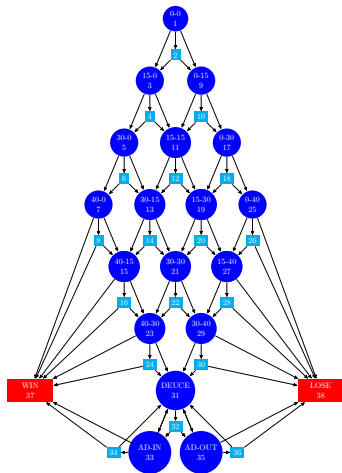
“Optimal rules for patent researchers”



Dube, Hitsch, Chintagunta (2010), *Marketing Science*

“Tipping and concentration in markets with indirect network effects”

Tennis is a Directional Dynamic Game

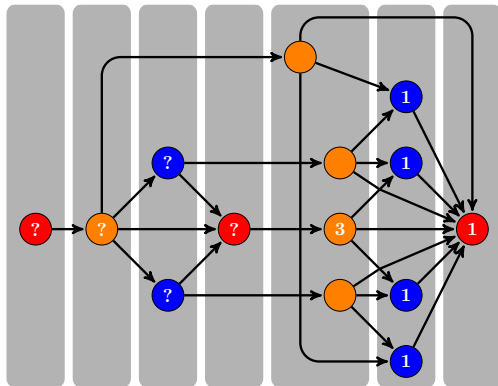


Anderson, Rosen, Rust, Wong (2024) *JPE*
“Disequilibrium Play in Tennis”

Multiplicity of stage equilibria

Number of equilibria in the higher stages depends on the selected equilibria

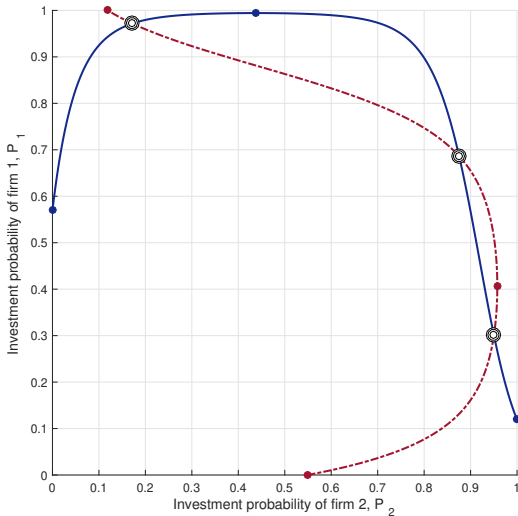
- ▶ State recursion proceeds **conditional on** equilibrium selection rule
- ▶ Multiplicity of stage equilibria $\Leftrightarrow$ multiplicity
- ▶ Can systematically combine different stage equilibria



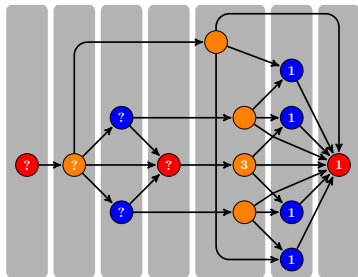
Best response functions

Typically one or three stage equilibria, but may be 5

- Smooth best response function with $\eta > 0$



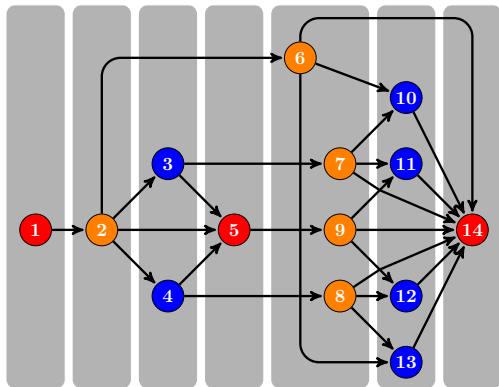
Multiplicity of stage equilibria $\iff$ Multiplicity of MPE



- ▶ State recursion proceeds **conditional on** equilibrium selection rule
- ▶ Selected equilibrium at downstream stage affects the equilibria and **number** of equilibria at upstream stages
- ▶ Need to systematically combine different stage equilibria $\rightarrow$ **Recursive Lexicographical Search** $\equiv$ **Depth-first tree traversal**

Indexing of points in the state space

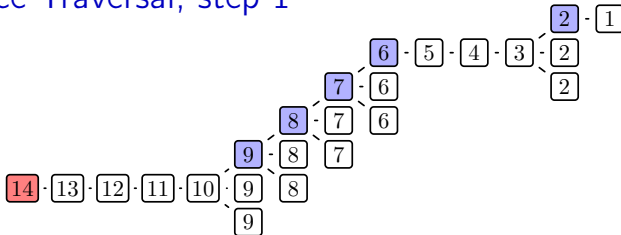
Lower index for dependent points, highest for terminal stage



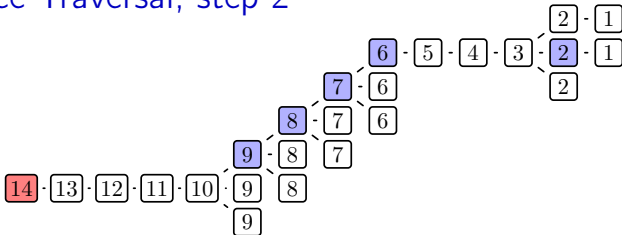
RLS = depth-first tree traversal (RLS tree on next slides)

- ▶ Root of the tree is one of the absorbing states
- ▶ Levels correspond to ordered stages (state points here)
- ▶ Branching happens when stages have multiple equilibria
- ▶ MPE of the game is given by a path from root to a leaf

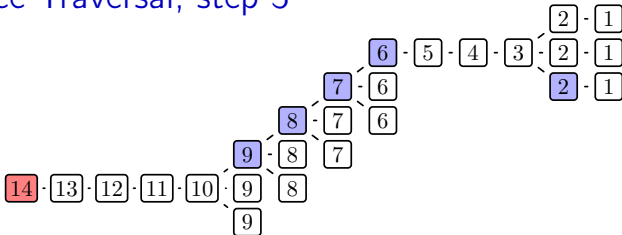
RLS Tree Traversal, step 1



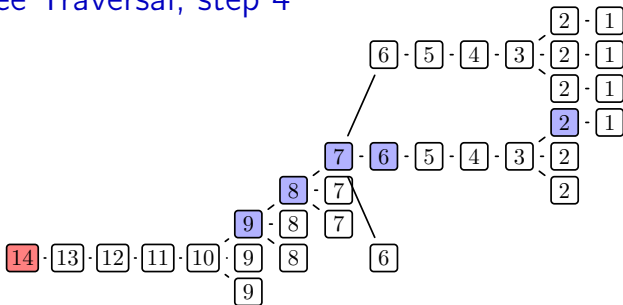
RLS Tree Traversal, step 2



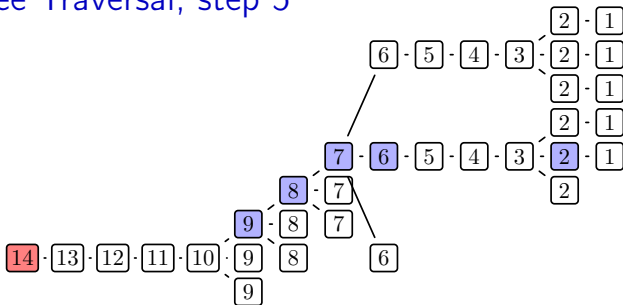
RLS Tree Traversal, step 3



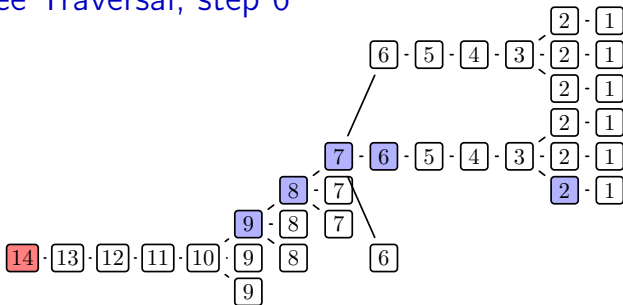
RLS Tree Traversal, step 4



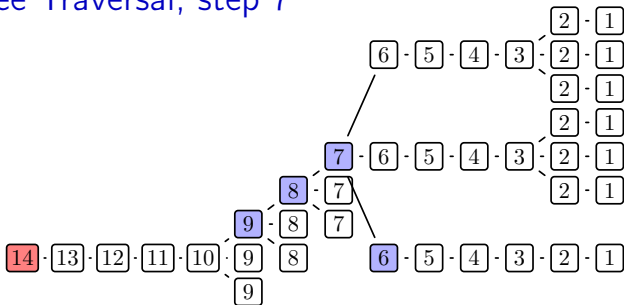
RLS Tree Traversal, step 5



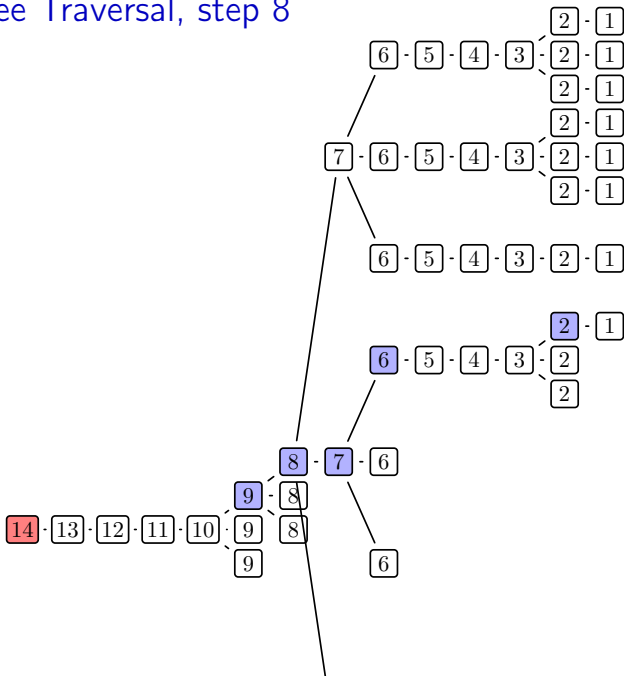
RLS Tree Traversal, step 6



RLS Tree Traversal, step 7



RLS Tree Traversal, step 8



Recursive Lexicographic Search (RLS) algorithm

Theorem (RLS theorem)

Assume there exists an algorithm that can find all MPE of every stage game of the DDG, and that the number of these equilibria is finite in every stage game.

Then the RLS algorithm finds all MPE of the DDG in a finite number of steps, which equals the total number of MPE.



Iskhakov, Rust and Schjerning, 2016, ReStud

“Recursive lexicographical search: Finding all markov perfect equilibria of finite state directional dynamic games.”

Nested Recursive Lexicographical Search (NRLS)

- Data from M independent markets from T periods, N players

$$Z = \{a^{ijt}, x^{it}\}_{i \in \{1 \dots M\}, j \in \{1, \dots, N\}, t \in \{1, \dots, T\}}$$

- Set of equilibria $\mathcal{E}(\theta) = \{V_\theta^k, P_\theta^k\}_{k \in \{1, \dots, K(\theta)\}}$

1. **Outer loop** Maximize likelihood over structural parameter θ

$$\theta^{ML} = \arg \max_{\theta \in \Theta} \mathcal{L}(Z, \theta)$$

2. **Inner loop** Maximize likelihood over equilibrium choice
 $k \in \{1, \dots, K(\theta)\}$

$$\mathcal{L}(Z, \theta) = \max_{k \in \{1, \dots, K(\theta)\}} \mathcal{L}(Z, \theta, P_\theta^k)$$

- With multiple equilibria in the data $\mathcal{L}(Z, \theta)$ has more elaborate form

Likelihood over the state space

Can efficiently represent likelihood by counts of observations

- ▶ With equilibrium k choice probabilities $P_j^k(a|x; \theta)$, likelihood is

$$\mathcal{L}(Z, \theta, P_\theta^k) = \sum_{t=1}^T \sum_{i=1}^M \sum_{j=1}^N \log P_j^k(a^{ijt} | x^{it}; \theta)$$

- ▶ Let ι index points in the state space
 $\iota = 1$ in the initial subset, $\iota = |X|$ in the terminal subset of X
- ▶ Denote n_ι the number of observations in state x_ι and $n_\iota^{a_j}$ the number of observations of player i taking action a_j at x_ι

$$n_\iota = \sum_{t=1}^T \sum_{i=1}^M \mathbb{1}\{x^{it} = x_\iota\} \quad n_\iota^{a_j} = \sum_{t=1}^T \sum_{i=1}^M \mathbb{1}\{a^{ijt} = a_j, x^{it} = x_\iota\}$$

- ▶ Then equilibrium-specific likelihood is given by

$$\mathcal{L}(Z, \theta, P_\theta^k) = \sum_{\iota=1}^{|X|} \sum_{j=1}^N \sum_{a_j} n_\iota^{a_j} \log P_j^k(a_j | x_\iota; \theta)$$

Branch and bound solution method



Land and Doig, 1960 *Econometrica*

- ▶ Old method for solving integer programming problems
- ▶ **Branching**: RLS tree
- ▶ **Bounding**: The bound function is partial likelihood of equilibrium k calculated on the subset of states $\iota \in \mathcal{S} \subset X$

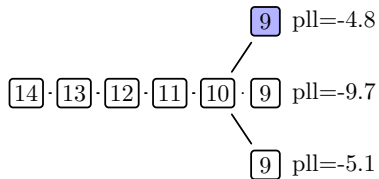
$$\mathcal{L}^{\text{part}}(\mathbf{Z}^{\mathcal{S}}, \theta, P_{\theta}^k) = \sum_{\iota \in \mathcal{S}} \sum_{j=1}^N \sum_{a_j} n_{\iota}^{a_j} \log P_j^k(a_j | x_{\iota}; \theta)$$

- ▶ Where $\mathbf{Z}^{\mathcal{S}} = \{(a, x) : x \in \mathcal{S}\}$ denotes data observed on $\mathcal{S}$
- ▶ Monotonic decreasing in cardinality of $\mathcal{S}$
(declines as more data is added)
- ▶ Equals to the full log-likelihood on the full state space when $\mathbf{Z}^{\mathcal{S}} = \mathbf{Z}$
(at the leaf of RLS tree, next slide)

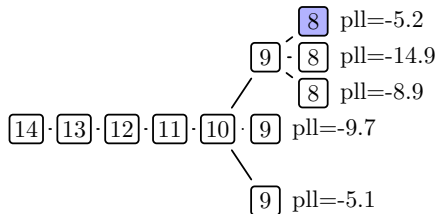
BnB on RLS tree, step 1

$\boxed{14} \cdot \boxed{13} \cdot \boxed{12} \cdot \boxed{11} \cdot \boxed{10}$ Partial loglikelihood = -3.2

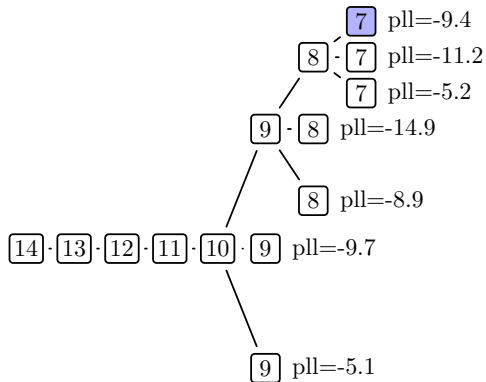
BnB on RLS tree, step 2



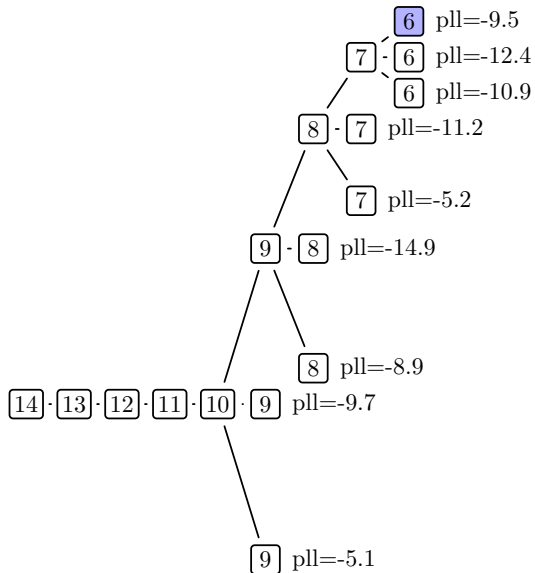
BnB on RLS tree, step 3



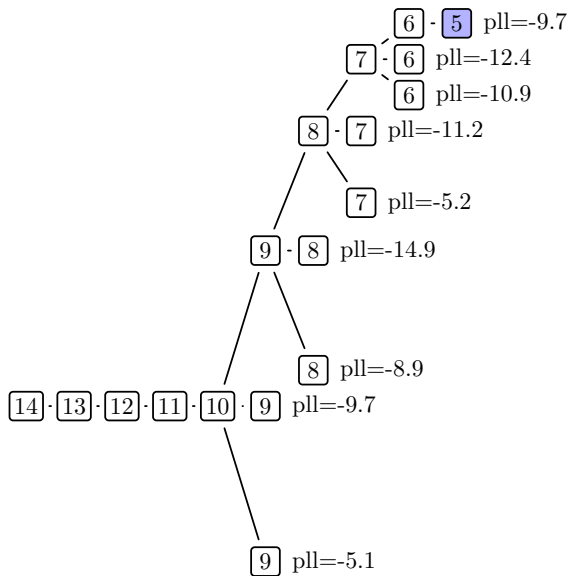
BnB on RLS tree, step 4



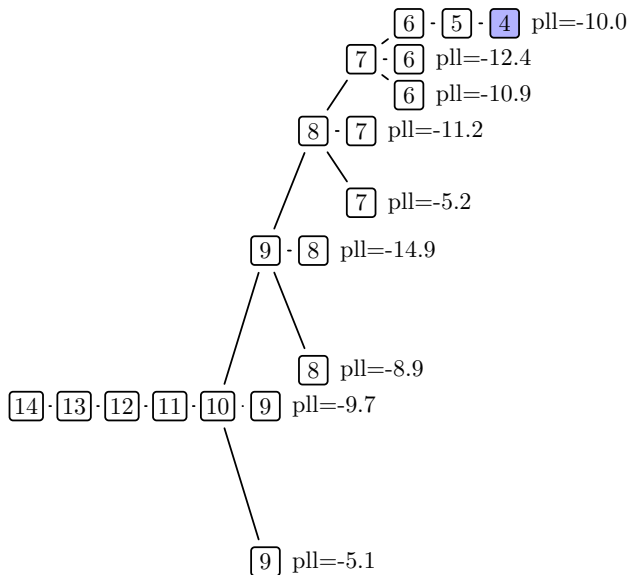
BnB on RLS tree, step 5



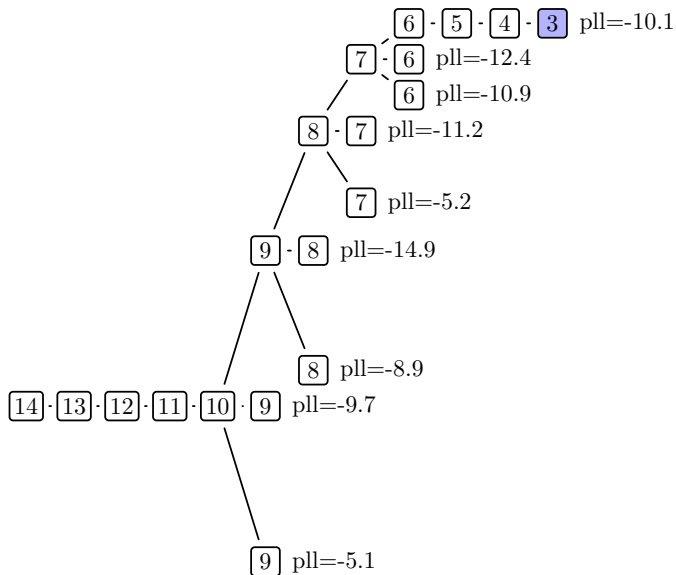
BnB on RLS tree, step 6



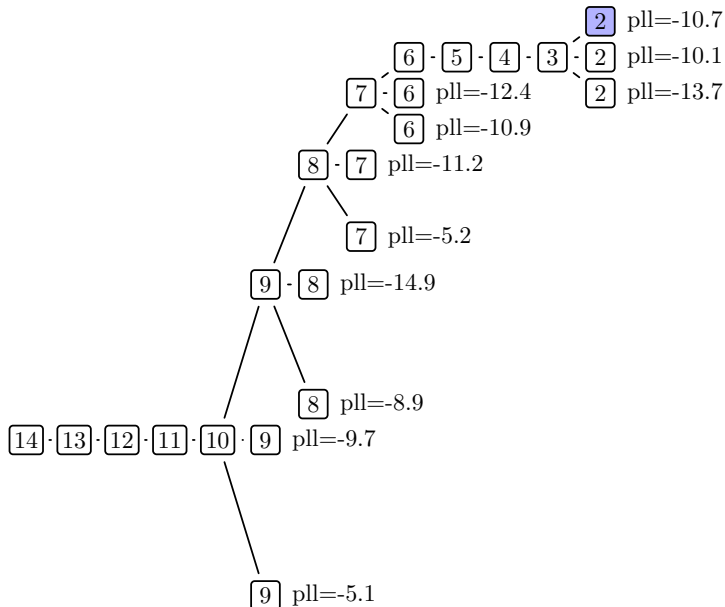
BnB on RLS tree, step 7



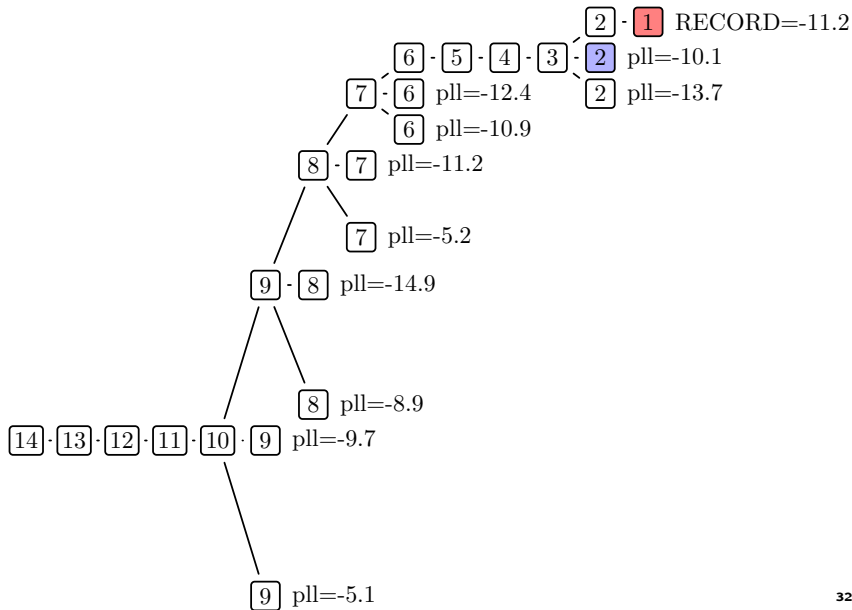
BnB on RLS tree, step 8



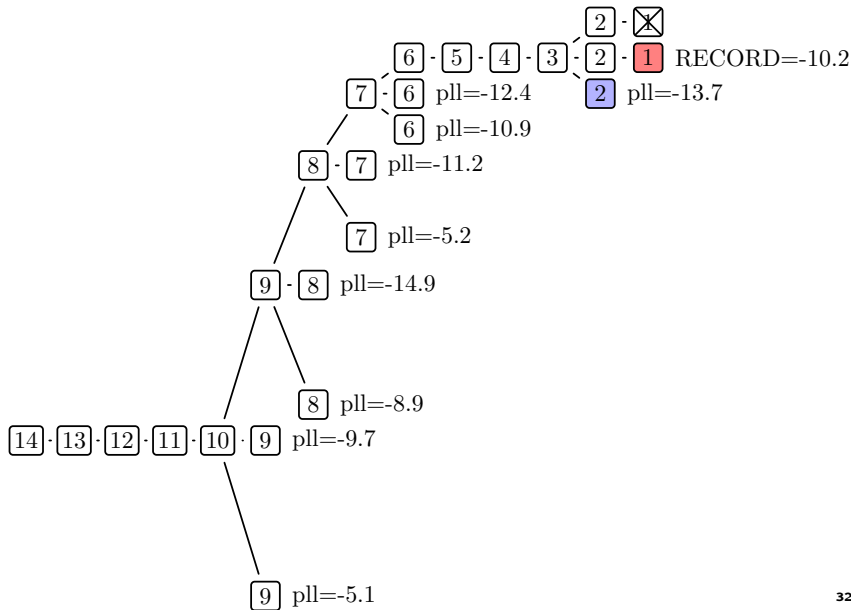
BnB on RLS tree, step 9



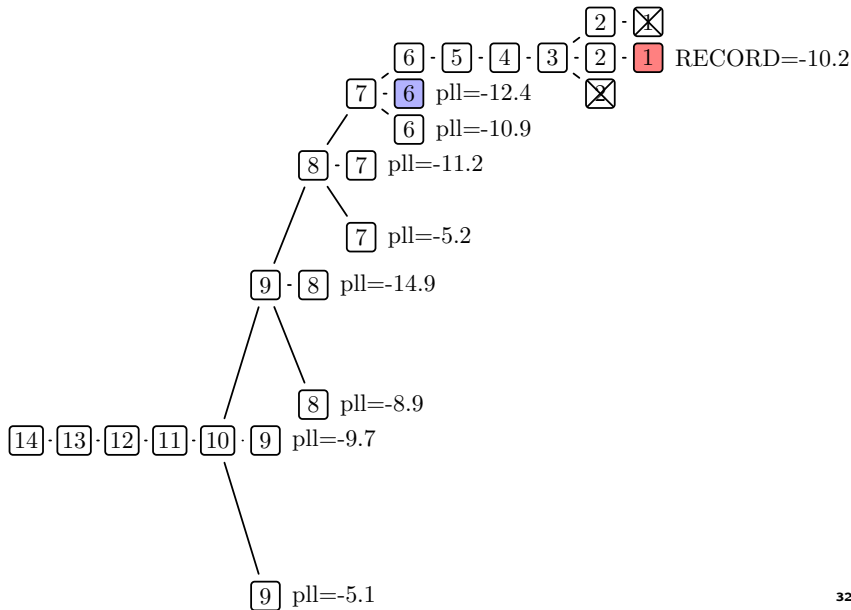
BnB on RLS tree, step 10



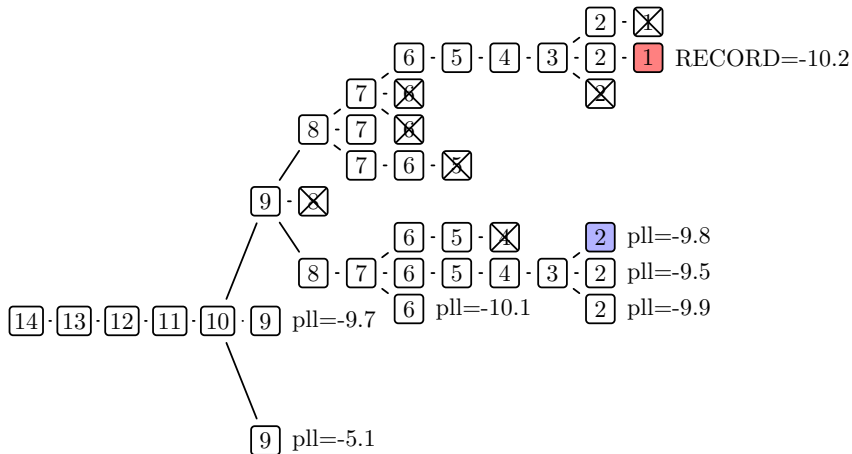
BnB on RLS tree, step 11



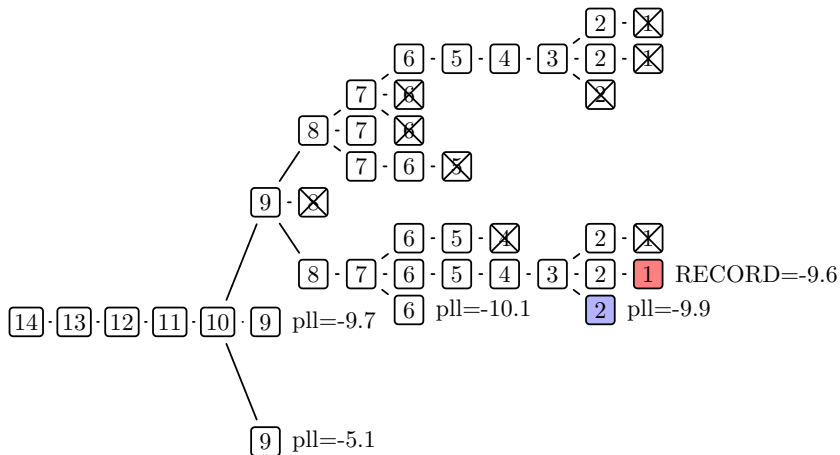
BnB on RLS tree, step 12



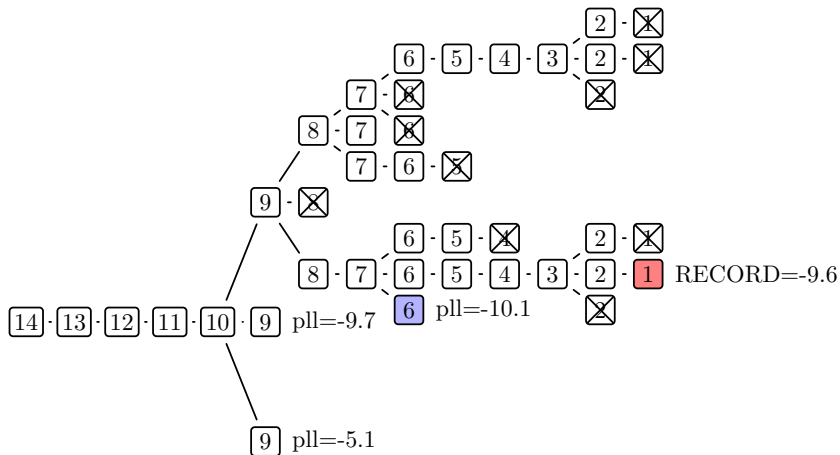
BnB on RLS tree, step 28



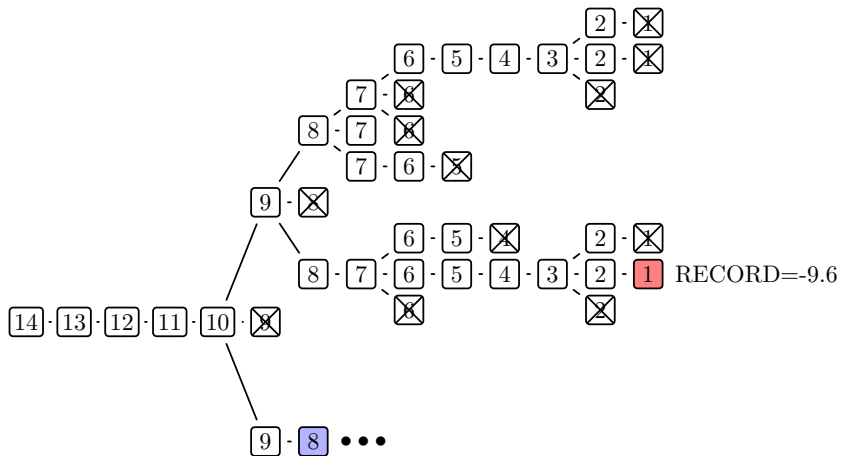
BnB on RLS tree, step 30



BnB on RLS tree, step 31



BnB on RLS tree, step 34



Refinement: non-parametric likelihood bounding

Much more powerful bound for BnB based on empirical frequencies

- ▶ Replace choice probabilities $P_j^k(a_j|x_\ell; \theta)$ with frequencies $n_\ell^{a_j}/n_\ell$

$$\mathcal{L}^{\text{non-par}}(Z^{\mathcal{S}}) = \sum_{\ell \in \mathcal{S}} \sum_{i=1}^J \sum_a n_\ell^{a_i} \log(n_\ell^a / n_\ell)$$

- ▶ $\mathcal{L}^{\text{non-par}}(Z^{\mathcal{S}})$ depends only on the counts from the data!
- ▶ Not hard to show algebraically that for any $Z^{\mathcal{S}}$ ($\approx$ Gibbs inequality)

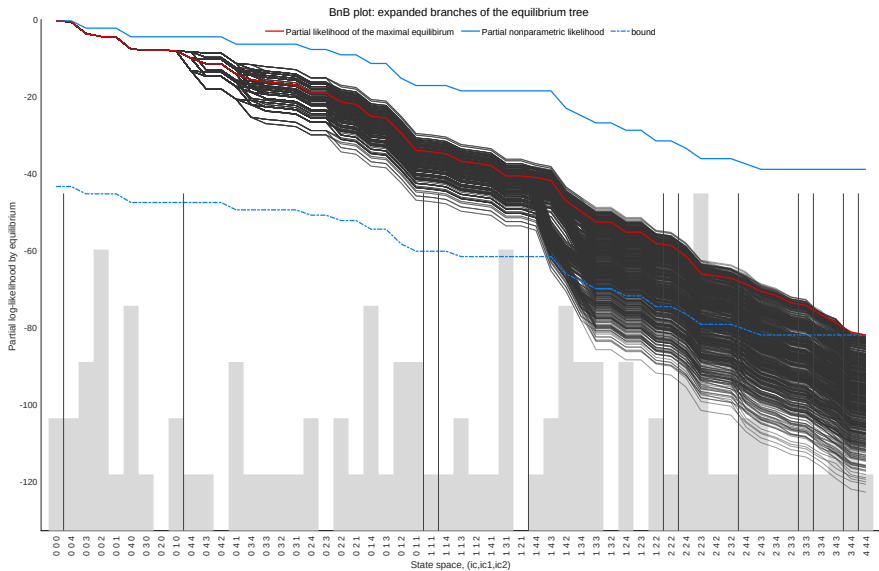
$$\mathcal{L}^{\text{non-par}}(Z^{\mathcal{S}}) > \mathcal{L}^{\text{part}}(Z^{\mathcal{S}}, \theta, P_\theta^k) \quad \forall \theta, k$$

- ▶ Therefore partial likelihood can be optimistically extrapolated by empirical likelihood at any step ℓ of the RLS tree traversal

$$\mathcal{L}^{\text{part}}(Z^{\{|X|, \dots, \ell\}}, \theta, P_\theta^k) + \mathcal{L}^{\text{non-par}}(Z^{\{\ell-1, \dots, 1\}})$$

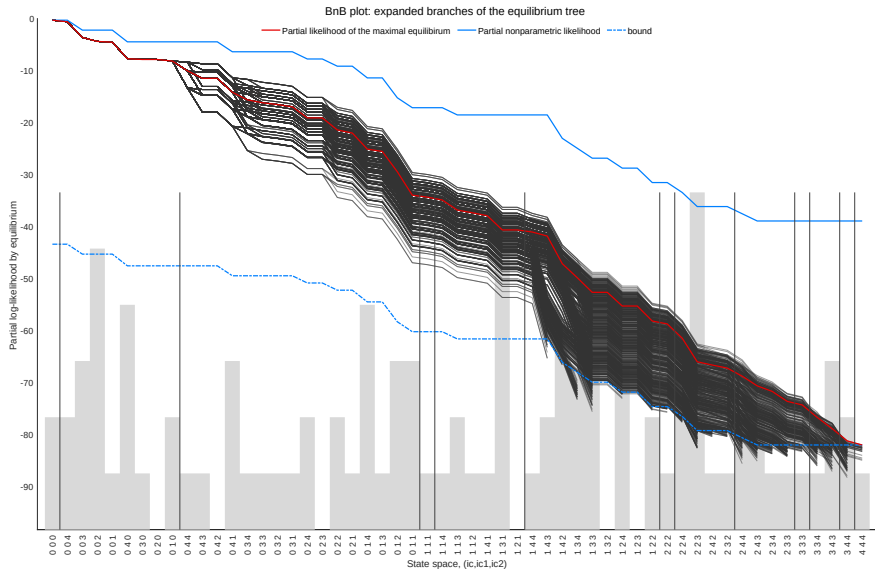
Nonparametric likelihood bounding

$\iota = |X| = 14$ (terminal state) on the left, $\iota = 1$ (initial state) on the right



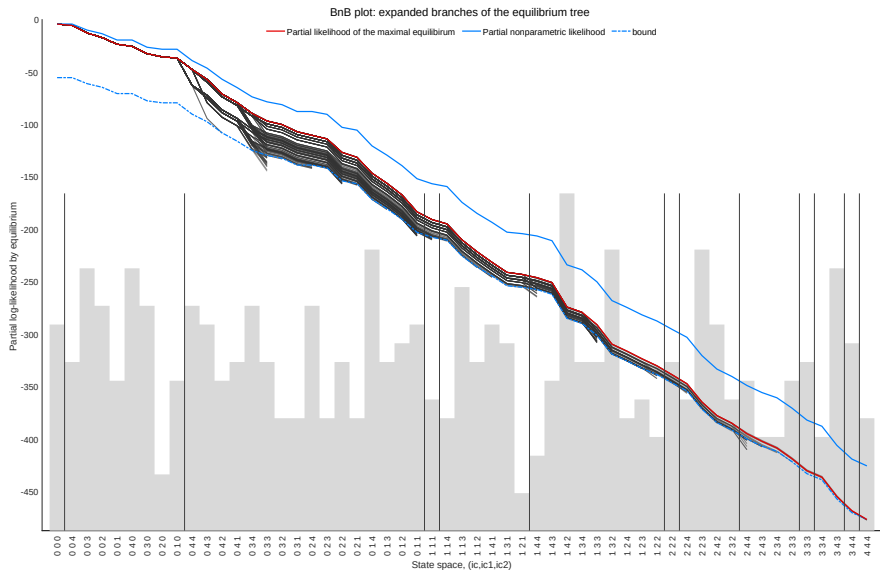
BnB with nonparametric likelihood bound

Greedy traversal + nonparametric likelihood bound



BnB with nonparametric likelihood bound, larger sample

Non-parametric $\rightarrow$ parametric likelihood as $N \rightarrow \infty$ at true $\theta \Rightarrow$ even less computation



BnB refinement with non-parametric likelihood

- ▶ For any amount of data the non-parametric likelihood is greater or equal to the parametric likelihood *algebraically*
- ▶ BnB augmented with a nonparametric likelihood bound gives sharper bounding rules $\rightarrow$ less computation
- ▶ With more data as $M \rightarrow \infty$ with T fixed
- ▶ Non-parametric log-likelihood converges to the actual likelihood
- ▶ The width of the band between the blue lines in the plots decreases
 - $\rightarrow$ Even sharper Bounding Rules
 - $\rightarrow$ Even less computation

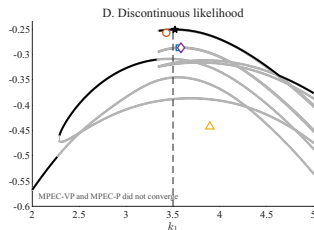
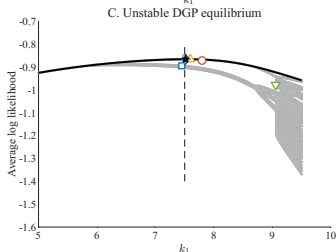
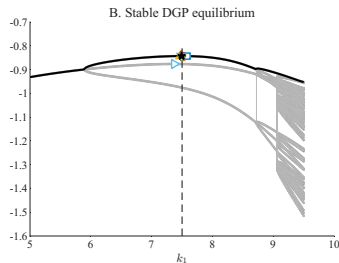
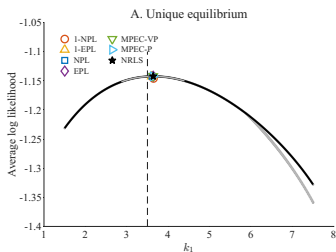
BnB yields exact solution of the inner integer maximization problem
 $\Rightarrow$ **MLE for any sample size, but easier to compute with more data!**

Monte Carlo: from uniqueness to 19,683 MPEs

	Experiment	States	MPEs	M	T
A	Unique equilibrium	30	1	1,000	5
B	Stable DGP equilibrium	30	3	5,000	5
C	Unstable DGP equilibrium	30	3	5,000	5
D	Discontinuous likelihood	55	9	1,000	5
E	Massive multiplicity	91	2,455	1,000	5
F	Convergence: smooth / discontinuous	30 / 55	3 / 9	50–5,000	5
G ₁	Five assignments, three distinct MPEs	55	81	5,000 [†]	5
G ₂	25 assignments, 23 distinct MPEs	91	19,683	5,000 [†]	10

- ▶ Same leapfrogging investment game; estimate the investment cost k_1
- ▶ Compare 1-NPL, 1-EPL, NPL, EPL, MPEC-VP, MPEC-P, and NRLS
- ▶ Read two diagnostics jointly: the residual asks “is it an equilibrium?”; the likelihood shortfall asks “is it the best-fitting equilibrium?”
- ▶ Experiments G_1 and G_2 : M is markets per **known group**[†]

A–D. The full-solution likelihood is an upper envelope



Light curves: equilibrium-specific likelihoods; black curve: full-solution envelope; dashed line: k_1^0 . A local estimator can lie on a valid branch but below the envelope; D shows the maximizing branch switching abruptly.

A. With a unique equilibrium, the estimators largely agree

Simultaneous moves; $n = 4$; 30 states; $k_1^0 = 3.50$; 1 MPE; $M = 1,000$; $T = 5$.

	1-NPL	1-EPL	NPL	EPL	MPEC-VP	MPEC-P	NRLS
Mean ($k_1^0 = 3.50$)	3.5279	3.4993	3.4971	3.4949	3.4949	3.4949	3.4949
Bias	0.0279	-0.0007	-0.0029	-0.0051	-0.0051	-0.0051	-0.0051
MCSD	0.1004	0.0709	0.0652	0.0704	0.0704	0.0708	0.0704
LL shortfall	—	—	-0.050	0.000	0.000	0.000	0.000
KL divergence	0.03254	0.00044	0.00021	0.00024	0.00024	0.00024	0.00024
$\ P - P_0\ $	0.11270	0.00995	0.00469	0.00495	0.00495	0.00500	0.00495
$\ \Psi(P; \theta) - P\ $	0.1618	0.0075	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$
Converged	100	100	100	100	100	99	100
CPU time (s)	0.01	0.03	2.00	0.13	0.27	0.23	3.81

- ▶ 1-NPL retains preliminary-CCP error: its equilibrium residual is 0.1618; iteration or constraints remove it.
- ▶ With only one branch, EPL, MPEC, and NRLS deliver essentially the same likelihood and estimate.
- ▶ Benign benchmark: local computation and full-solution MLE agree because selection does not bind.

B. MPEC can find an equilibrium and still miss the MLE

Simultaneous moves; $n = 4$; 30 states; $k_1^0 = 7.50$; 3 MPEs; stable DGP equilibrium ($\rho_\psi = 0.778$); $M = 5,000$; $T = 5$.

	1-NPL	1-EPL	NPL	EPL	MPEC-VP	MPEC-P	NRLS
Mean ($k_1^0 = 7.50$)	7.5516	7.5052	7.4984	7.4992	7.6532	7.3512	7.4992
Bias	0.0516	0.0052	-0.0016	-0.0008	0.1532	-0.1488	-0.0008
MCSD	0.1788	0.0511	0.0606	0.0341	0.9974	0.4714	0.0341
LL shortfall	—	—	-1.2	0.000	-1062.7	-776.8	0.000
KL divergence	0.02557	0.00159	0.00040	0.00013	0.23536	0.16051	0.00013
$\ P - P_0\ $	0.11085	0.01749	0.00490	0.00280	0.17466	0.20957	0.00280
$\ \Psi(P; \theta) - P\ $	0.1709	0.0157	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$
Converged	100	100	100	100	98	97	100
CPU time (s)	0.01	0.03	1.58	0.12	0.76	1.23	4.22

- ▶ The large sample and stable DGP give accurate starting CCPs: NPL, EPL, and NRLS perform well.
- ▶ MPEC finds a valid equilibrium (residual $< 10^{-6}$) but a disastrous local solution—a local minimum of negative log-likelihood: shortfalls -1062.7 and -776.8.
- ▶ **Feasibility is not global optimality:** the near-zero residual certifies an equilibrium, not the full-solution MLE.

C. Instability exposes the limits of local iteration

Same game as B, but the DGP equilibrium is unstable ($\rho_\psi = 1.211$): $n = 4$; 30 states; $k_1^0 = 7.50$; 3 MPEs; $M = 5,000$; $T = 5$.

	1-NPL	1-EPL	NPL	EPL	MPEC-VP	MPEC-P	NRLS
Mean ($k_1^0 = 7.50$)	7.5424	7.4553	7.3928	7.4804	7.7313	7.6310	7.5018
Bias	0.0424	-0.0447	-0.1072	-0.0196	0.2313	0.1310	0.0018
MCSD	0.1714	0.2221	0.0561	0.1580	0.7299	0.8987	0.0382
LL shortfall	—	—	-765.2	-11.4	-502.1	-920.6	0.000
KL divergence	0.02271	0.00784	0.15996	0.00257	0.11452	0.20182	0.00012
$\ P - P_0\ $	0.09757	0.02547	0.20709	0.00619	0.03860	0.02504	0.00307
$\ \Psi(P; \theta) - P\ $	0.1601	0.0149	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$
Converged	100	100	18	100	99	98	100
CPU time (s)	0.01	0.03	1.95	0.14	0.61	0.70	3.88

- ▶ Holding B's model fixed isolates stability: policy iteration is repelled from the statistically relevant branch; NPL converges only 18 times.
- ▶ EPL fixes much of the local stability problem, but still averages 11.4 log-likelihood units below the best valid solution.
- ▶ MPEC again satisfies the constraints yet reaches inferior branches; NRLS alone compares the complete equilibrium correspondence.

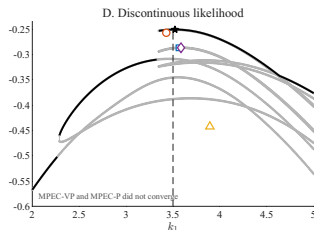
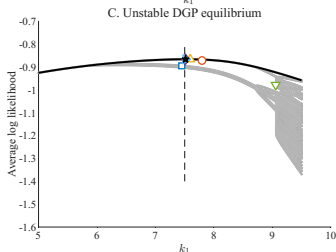
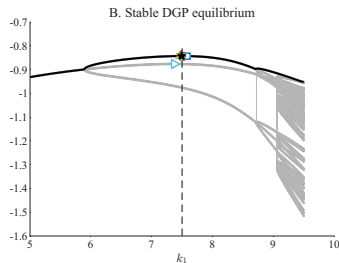
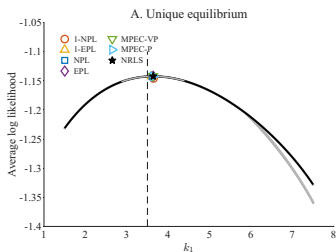
D. Discontinuous likelihood correspondence

Alternating moves; $n = 5$; 55 states; $k_1^0 = 3.50$; 9 MPEs; unstable DGP near a likelihood cliff; $M = 1,000$; $T = 5$.

	1-NPL	1-EPL	NPL	EPL	MPEC-VP	MPEC-P	NRLS
Mean ($k_1^0 = 3.50$)	3.4974	3.6219	3.5514	3.6477	3.6594	3.6703	3.5021
Bias	-0.0026	0.1219	0.0514	0.1477	0.1594	0.1703	0.0021
MCSD	0.1400	0.2117	0.0713	0.1290	0.1269	0.1158	0.0325
LL shortfall	—	—	-219.4	-222.1	-271.0	-267.5	0.000
KL divergence	0.01512	0.05395	0.04889	0.04495	0.04102	0.04078	0.00016
$\ P - P_0\ $	0.62850	0.77118	0.86124	0.83062	0.66562	0.65879	0.01610
$\ \Psi(P; \theta) - P\ $	0.7638	0.3628	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	1.5×10^{-6}
Converged	100	100	100	100	28	27	100
CPU time (s)	0.02	0.04	0.85	0.22	2.30	0.52	22.0

- ▶ NPL and EPL converge and satisfy equilibrium restrictions, yet repeatedly track the wrong side of the likelihood cliff.
- ▶ MPEC is both locally trapped and numerically fragile: only 28 and 27 successful replications.
- ▶ NRLS tracks the globally maximizing branch when the optimizer over equilibria changes discontinuously.

A–D. The full-solution likelihood is an upper envelope



Light curves: equilibrium-specific likelihoods; black curve: full-solution envelope; dashed line: k_1^0 . A local estimator can lie on a valid branch but below the envelope; D shows the maximizing branch switching abruptly.

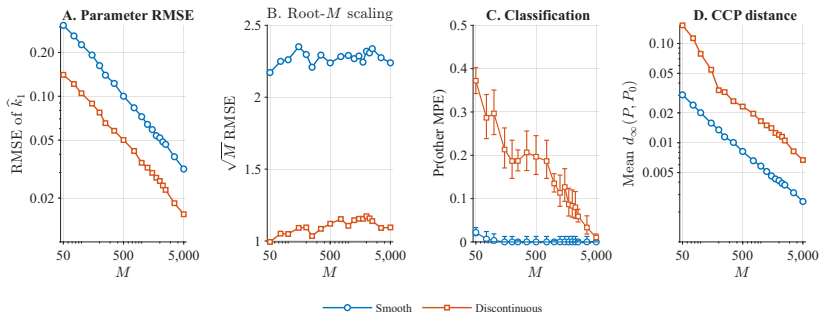
E. NRLS remains feasible with 2,455 equilibria

Simultaneous moves; $n = 6$; 91 states; $k_1^0 = 3.75$; 2,455 MPEs; one equilibrium played; $M = 1,000$; $T = 5$.

	1-NPL	1-EPL	NPL	EPL	MPEC-VP	MPEC-P	NRLS
Mean ($k_1^0 = 3.75$)	3.7096	2.7906	3.7127	3.7891	—	—	3.7424
Bias	-0.0404	-0.9594	-0.0373	0.0391	—	—	-0.0076
MCSD	0.1109	0.7214	0.0681	0.4072	—	—	0.0303
LL shortfall	—	—	-66.5	-467.6	—	—	0.000
KL divergence	∞	∞	14.075	1.22×10^4	—	—	0.32429
$\ P - P_0\ $	0.82204	0.87083	0.65580	0.79241	—	—	0.07454
$\ \Psi(P; \theta) - P\ $	0.9636	0.8095	$< 10^{-6}$	$< 10^{-6}$	—	—	6.1×10^{-6}
Converged	100	100	18	68	—	—	100
CPU time (s)	0.16	0.39	11.3	4.01	—	—	4.73

- ▶ Sequential methods deteriorate sharply: NPL converges 18 times, EPL 68, and both often select poorly fitting equilibria.
- ▶ Blank MPEC entries are substantive: even sparse analytical constraints are infeasible at this dimension.
- ▶ Exhaustive RLS once takes 10 minutes 39 seconds; NRLS averages 4.73 seconds without enumerating all MPEs at every trial value.

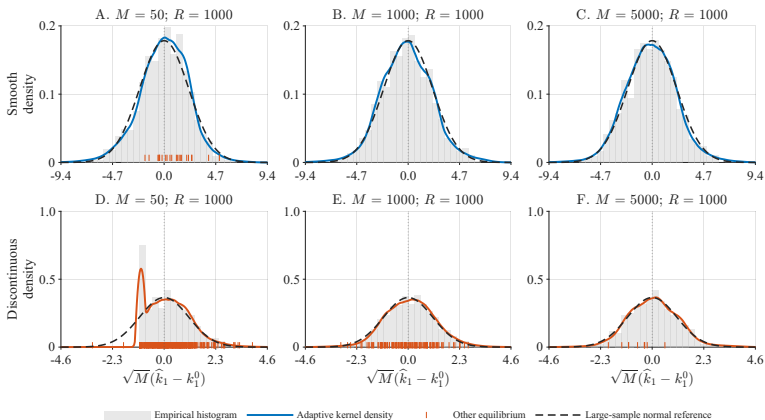
F. As M grows, $\hat{\theta}_{NRLS}$ and equilibrium selection converge



Repeat B (smooth) and D (discontinuous); $M = 50$ –5,000; $T = 5$.

- ▶ Parameter RMSE follows the root- M pattern in both designs.
- ▶ Near the cliff, selecting another MPE falls from 37.2% at $M = 50$ to 1.0% at $M = 5,000$.
- ▶ Discrete classification becomes reliable before continuous parameter uncertainty disappears: the finite-sample analogue of oracle equivalence.

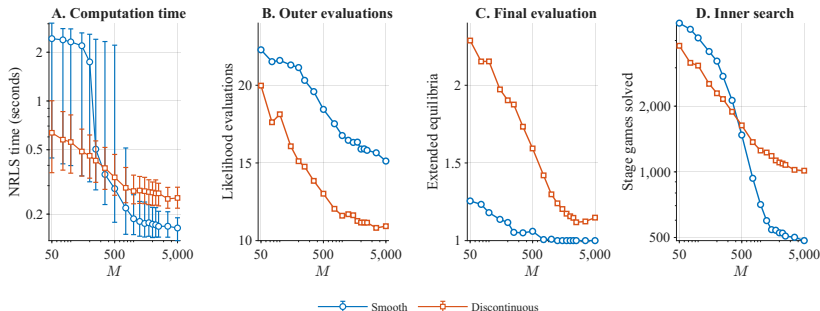
F. The likelihood cliff creates a finite-sample distortion



Repeat B (smooth) and D (discontinuous); $M = 50\text{--}5,000$; $T = 5$.

- ▶ As M grows, those events vanish and $\sqrt{M}(\hat{k}_1 - k_1^0)$ approaches the normal oracle reference.
- ▶ A likelihood cliff creates finite-sample distortion; it does not by itself imply nonstandard asymptotics away from an existence-set boundary.

F. More data reduce the computational burden of NRLS



Repeat B (smooth) and D (discontinuous); $M = 50-5,000$; $T = 5$.

- ▶ Median NRLS time falls even though each likelihood evaluation processes more observations.
- ▶ More data sharpen the bound: fewer outer evaluations, almost one fully extended final branch, and fewer stage games solved.
- ▶ Information is a computational resource.

Multiple equilibria in the data: known groups

G_1 : five assignments

- ▶ Simultaneous moves;
 $n = 5$; 55 states
- ▶ $k_1^0 = 9.25$
- ▶ 81 MPEs
- ▶ Five groups;
three distinct MPEs played
- ▶ 5,000 markets/group; $T = 5$

G_2 : 25 assignments

- ▶ Simultaneous moves;
 $n = 6$; 91 states
- ▶ $k_1^0 = 9.50$
- ▶ 19,683 MPEs
- ▶ 25 groups;
23 distinct MPEs played
- ▶ 5,000 markets/group; $T = 10$

- ▶ Common structural parameter; NRLS selects an equilibrium separately for each known group.
- ▶ Pooled NPL/EPL impose one common CCP profile and are benchmarks for the grouped criterion.
- ▶ Latent group membership is not estimated in these experiments.

G_1 . NRLS recovers every one of the five assignments

Simultaneous moves; $n = 5$; 55 states; $k_1^0 = 9.25$; 81 MPEs; 5 known groups, 3 distinct MPEs; $M = 5,000/\text{group}$; $T = 5$.

	1-NPL (pooled)	NRLS (grouped)
Mean ($k_1^0 = 9.25$)	9.2099	9.2545
Bias	-0.0401	0.0045
MCSD	0.1502	0.0411
LL shortfall	—	0.000
KL divergence	0.32943	0.00028–0.00040
$\ P - P_0\ _\infty$	0.32787	0.00240–0.00287
$\ \Psi(P; \theta) - P\ _\infty$	0.4609	$< 10^{-6}$
Converged	100	100
CPU time (s)	0.02	20.7

- ▶ Every group assignment is recovered in every replication.
- ▶ The first three groups play the same equilibrium, as correctly identified by NRLS.
- ▶ The pooled 1-NPL benchmark cannot represent group-specific play: large KL error and residual 0.4609.

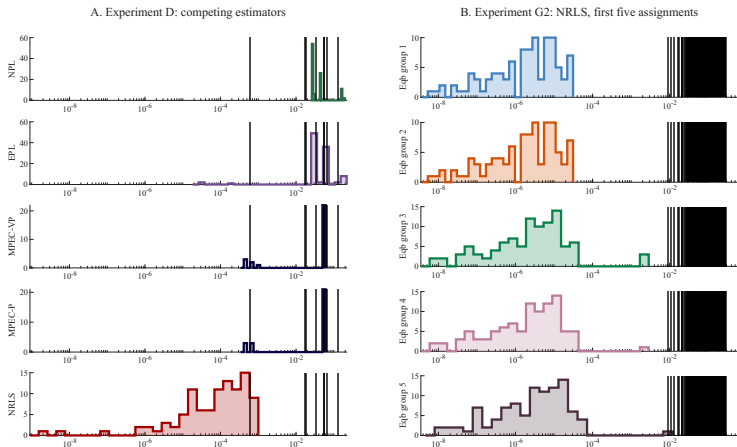
G_2 . With 19,683 MPEs, estimation beats enumeration

Simultaneous moves; $n = 6$; 91 states; $k_1^0 = 9.50$; 19,683 MPEs; 25 known groups, 23 distinct MPEs; $M = 5,000/\text{group}$; $T = 10$.

	1-NPL	1-EPL	NPL	EPL	NRLS
Mean ($k_1^0 = 9.50$)	9.5392	9.1207	—	9.4090	9.4999
Bias	0.0392	-0.3793	—	-0.0910	-0.0001
MCSD	0.0286	0.1600	—	0.0559	0.0041
LL shortfall	—	—	—	-543.7	0.000
KL divergence	0.05753	0.10384	—	0.05045	4.67×10^{-6} – 9.71×10^{-5}
$\ P - P_0\ _\infty$	0.18517	0.26187	—	0.26430	0.00059–0.00572
$\ \Psi(P; \theta) - P\ _\infty$	0.1870	0.1038	—	$< 10^{-6}$	$< 10^{-6}$
Converged	100	100	0	100	100
CPU time (s)	0.05	0.12	—	1.04	189.4

- ▶ NRLS recovers all 25 assignments in all 100 replications.
- ▶ Exhaustive RLS once: **2 hours 27 minutes**; complete NRLS estimation: **189 seconds**.
- ▶ More observations sharpen the bound enough to preserve the exact inner maxima while pruning aggressively.
- ▶ NRLS group KL errors are below 10^{-4} ; the closest alternative MPE is about 10^{-2} away.

D and G_2 . Equilibrium-selection errors are meaningful



- ▶ In D, local-method errors are repeated, economically meaningful selections—not a few innocuous outliers.
- ▶ In G_2 , NRLS errors lie orders of magnitude below the vertical lines marking alternative MPEs.

NRLS estimator for directional dynamic games

Full-solution MLE requires maximization over a large finite set of MPE → branch-and-bound with a likelihood upper bound

1. Each stage game → non-linear solver, **specific to the model**
 2. Combining stage game solutions to full game MPEs → **State Recursion algorithm**
 3. Solving for all MPE → **Recursive Lexicographic Search**
 4. Structural estimation → **Nested** Recursive Lexicographic Search
- ▶ A small equilibrium residual or numerical convergence does **not** establish global credibility under multiplicity
 - ▶ BnB NRLS avoids full enumeration without changing the inner discrete optimum, conditional on a complete stage-game solver
 - ▶ The nonparametric likelihood bound sharpens with sample size → more data can mean less computation
 - ▶ Reliable under stable and unstable equilibria, likelihood cliffs, and massive multiplicity
 - ▶ Recovers heterogeneous equilibrium assignments when groups are known